PRODUCT MANAGEMENT CASEBOOK

A comprehensive guide to help you prepare for Product Management Roles

INSTITUTE OF RURAL MANAGEMENT ANAND

FOREWORD

The Product Management Casebook, curated by the Product Management Club of IRMA, embodies a distinctive approach to equipping students, emerging leaders, and practitioners with hands-on insights into the dynamic realm of product management. Crafted with rigour and creativity, this casebook is a **"COMPASS"** for navigating real-world challenges, blending theoretical frameworks with actionable strategies. I extend my heartfelt commendation to the dedicated members of the Product Management Club for their relentless efforts in assembling this invaluable resource, which reflects IRMA's commitment to fostering managerial excellence.

Prof. Janak Suthar
Mentor, Product Management Club
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Introduction To IRMA Product Management Case Book

We take great pride in presenting the first edition of the **Product Management Casebook**, a resource designed to help you excel in securing top product management and consulting roles.

Our goal with this case book is to bridge the gap between academic knowledge and its practical, real-world applications. As you prepare for your Summer Internship Segment (SIS) and Final placements, envision yourself as a Product Manager navigating dynamic challenges—this guide will equip you with the tools to apply theoretical principles and craft innovative, impactful solutions. However, this case book is just the beginning. Product Management is a constantly evolving field that requires curiosity, adaptability, and continuous learning. We encourage you to go beyond these pages—explore new perspectives, seek fresh insights, and infuse your personal creativity into the process. Success isn't just about having the right tools; it's about how you leverage them. So stay open-minded, keep learning, and trust your instincts. A heartfelt thank you to the dedicated **Product Management Team**, whose hard work and passion made this case book a reality. Their commitment embodies the essence of product management—collaboration, innovation, and the drive to create meaningful impact.

We understand that the journey to **SIS and Final Placements** can be demanding, but never let the pressure overshadow your passion for learning. Embrace the process, stay focused on your goals, and remember—every step brings you closer to mastering the art of product management.

Wishing you all the best in your journey, and may you shine bright in your SIS and final placements.

Product Management Club

2024-2025

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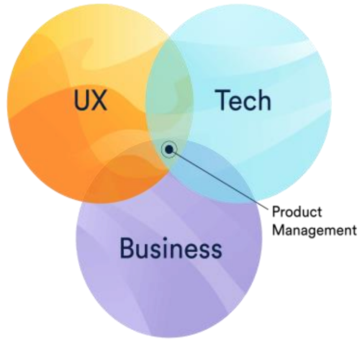
INTRODUCTION TO PRODUCT MANAGEMENT

What is Product Management?

Product Management (PM) is a strategic function within a company that focuses on developing, launching, and improving products that meet customer needs while aligning with business goals.

A Product Manager acts as the bridge between different teams like engineering, design, marketing, sales, and customer support to ensure the product is successful in the market and stays with the product throughout its lifecycle.



Key Responsibilities of a Product Manager:

- Conducting research, analysing data, and gathering feedback to identify pain points and opportunities.
- Setting a clear direction for the product based on market trends and business objectives.
- Deciding what features to build, balancing customer expectations with technical feasibility.
- Working closely with developers, designers, and marketers to ensure smooth execution.
- Overseeing the product's rollout, analysing performance, and iterating based on user feedback.

Skills required to be a good Product Manager

- Good organizational skills (in your thinking as well!)
- On point decision making skills
- Communication skills with not just the customers and but also with different teams

TYPES OF PRODUCT MANAGERS

Core Product Manager

Imagine you're at the heart of a product, shaping its very foundation. That is what a Core PM does. They are the architects of a product's key features and functionality, ensuring it solves real problems for users. Whether it is refining the way Google Drive handles file sharing or simplifying loan applications for rural farmers, they focus on building something valuable from the ground up. Success here is not about flashy marketing but creating a product that works & keeps users coming back.

Growth Product Manager

Now, let's talk about the Growth Product Manager—the person responsible for making sure people not only use a product but stay engaged. Ever wondered why you receive perfectly timed notifications from apps like Swiggy or how a telemedicine platform ensures rural users keep coming back? That's growth strategy in action. These PMs use A/B testing, analytics, and funnel optimization to improve user acquisition and retention. It's like fine-tuning an engine to make sure it runs as smoothly and efficiently as possible.

Technical Product Manager (TPM)

For those who love technology at its core, TPM is the ultimate role. They work on highly complex, backend-heavy products—things like cloud computing, AI, and cybersecurity. Think about an AWS PM designing next-gen cloud storage features or an Agri-tech PM developing AI-powered pest detection systems for farmers. If you have a technical background and love working on cutting-edge innovations, this is where you'd thrive.

Data Product Manager

Data Product Manager is the mastermind behind leveraging vast amounts of information to enhance decision-making. Imagine working at Netflix, fine-tuning recommendation algorithms to ensure every user gets the perfect movie suggestion. Or picture a rural microfinance platform using AI to determine creditworthiness for unbanked users. It is all about making smarter decisions through data analytics.

Platform Product Manager

The Platform Product Manager operates behind the scenes, building robust internal tools and infrastructure that power businesses. While they don't create consumer-facing features, their work is critical. Picture someone at Stripe refining payment processing APIs or a PM in rural fintech developing the backend system for UPI-based banking. Their work ensures that everything runs smoothly, securely, and at scale.

Consumer Product Manager

Now, if you're fascinated by consumer psychology and love crafting seamless experiences, the Consumer Product Manager role might be for you. These PMs design products for the masses, focusing on usability, engagement, and retention. Ever wondered how Instagram's Reels feature keeps you scrolling or how rural e-commerce platforms help farmers discover the right products? That is the magic of consumer product management.

Enterprise/B2B Product Managers

On the other hand, Enterprise/B2B Product Managers focus on building software for businesses rather than individual users. They create powerful tools that help companies streamline operations—like a

PM at Salesforce improving CRM features or someone in rural logistics developing supply chain software for agricultural markets. It's a high-stakes game where success means improving efficiency at scale.

Hardware Product Manager

Finally, for those passionate about tangible, real-world products, the Hardware Product Manager role is a perfect fit. These PMs oversee the development of physical products embedded with software. Think of a Tesla PM working on battery management systems or a rural electrification PM designing affordable solar power kits. Their job isn't just about the tech—it's about ensuring seamless integration between hardware and software.

Each of these roles comes with its own challenges and rewards. Some PMs work on front-end consumer experiences, others focus on infrastructure, and some dive deep into data or hardware. The key is to find the area that excites you the most. So, which one sparks your interest?

PRODUCT MANAGEMENT IN RURAL CONTEXT

In rural management, product management plays a crucial role in developing solutions tailored to rural markets, such as:

- Managing the development of mobile apps for farmers, precision farming tools, or digital marketplaces providing holistic Agri-tech solutions.
- Overseeing microfinance platforms, rural banking solutions, or insurance schemes designed for low-income communities.

In these cases, Product Managers must prioritize accessibility, affordability, and usability while ensuring the product meets local needs and infrastructure challenges.

CAREER TRAJECTORY IN PRODUCT MANAGEMENT

Associate Product Managers are responsible for prioritizing tasks with a defined set of constraints. They are not necessarily determining which tasks they are performing. Instead, they make scoping and prioritization decisions around the tasks or projects they are assigned.

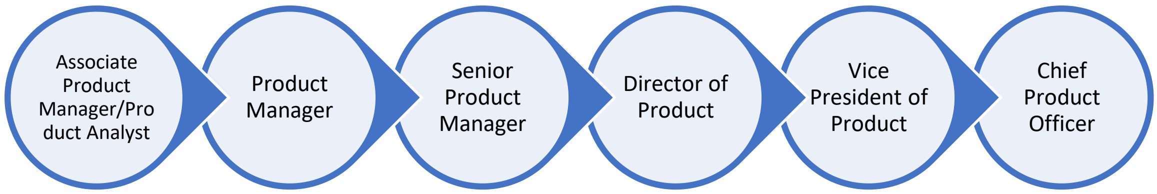
This mid-level product role is like the associate-level **Product Manager** position. Except that in addition to being the product's "go-to" resource for other teams, you will also be the point person for the product team related to your product.

At a minimum, a successful **Senior Product Manager** will come in with professional experience that demonstrates the ability to think on their feet that they can be accountable for decisions and make data-driven decisions based on many complex, interdependent factors.

A **Director**-level role in Product will require leadership experience. Other skills include excellent written and verbal communication, experience presenting to the executive leadership, and the ability to build and trust a team to do the work you previously did as a sole contributor.

At the level of **VP**, the product manager career path, you are significantly less involved with the product development process's hands-on activities.

Chief Product Officer is either an expanded variation of the VP of Product role or one that oversees many VPs of product rolling up to one product leader.



Source: ProductPlan

PREPARING FOR PM ROLES

All the PM roles are different which is why their interviews are also different. Companies tend to approach individual product management interviews differently depending on their product type and its strategy, industry, company size, culture, etc.

A few things you should keep in mind:

- The product manager interview processes are generally challenging likely to last several rounds.
- Although interview styles will vary, all companies are mostly look for curious, dedicated candidates who have problem solving attitude.
- Showing confidence as you share your knowledge and experience will help you feel your best in each conversation.

There are two main types of interviews you are likely to encounter:

- **Behavioural Interviews or HR Rounds:**
  These interviews focus on your past experiences, soft skills, and cultural fit within the organization. You will be asked questions about your background, previous job experience,

leadership abilities, and problem-solving skills. The goal is to understand how you have tackled challenges in the past and whether your approach aligns with the company's values and work culture. Expect to answer questions that start with "Tell me about a time when…" or "Give an example of how you handled…," covering areas like conflict resolution, teamwork, decision-making, and adaptability.

- **Case-style Interviews and Guesstimates:**
  These rounds test your analytical thinking, structured problem-solving, and creativity by presenting real-world business problems or hypothetical scenarios. You may be asked to solve a product design challenge, market-sizing question or a strategic business problem. The focus is on how logically you break down the problem, structure your thought process, and provide a well-reasoned solution. Guesstimates—where you estimate numerical values (e.g., "How many ATMs are there in Delhi?")—are commonly used to assess logical reasoning and quantitative thinking.

HOW TO APPROACH A PM INTERVIEW?

Knowing the Company by Heart

You should know everything about the company, from its product portfolio to the people interviewing you. Seek to understand the problems and pain points they want to solve. Know about its customers, competitors, suppliers, etc. For example, knowing whether its product is B2B or B2C can help you give a more relevant response about developing product strategy — because the way organizations serve these markets will differ. You should also work upon ways as to how you would approach pain points faced and improve the product features accordingly.

Leveraging your Experience

Preparing questions based on your skills and experience which align with the job description would help you drive the interview. This is especially valuable if you are breaking into entry-level product management roles or transitioning from another discipline. For example, if you have just graduated you should talk product-related courses or certifications done by you. If you are switching from a different role, you may want to highlight your skills and interests accordingly. Quantifiable results and accomplishments are always more impactful.

Demonstrating your values and personality which align with company's values and goals would give you an edge. Having eagerness to learn also goes a long way (even if you are experienced professional).

Get familiar with Typical Questions

Detailed product related topics and frameworks will be discussed later.

For now, these few simple interview frameworks might help you get an idea. Try practicing them.

DIGS	▪ **Dramatize the situation:** Emphasize the compelling details and craft a narrative. ▪ **Indicate the alternatives:** Explain how you arrived at the solution you chose. ▪ **Go through what you did:** Describe your efforts in detail. ▪ **Summarize your impact:** Show how your solution provided value to customers or the business.
EAST	▪ **Explain:** Describe the situation and its significance. ▪ **Approach:** Outline the approach you took, including research or frameworks. ▪ **Solution:** Detail the solution you ultimately implemented. ▪ **Trade-offs:** Discuss any trade-off decisions, challenges, or constraints you faced.
Rule of Threes	▪ Offer examples, stories, and other responses in lists of three to make them more memorable and engaging.
SPADE	▪ **Situation:** Briefly share the scenario. ▪ **Problem:** Define the core problem or challenge you faced. ▪ **Alternatives:** List the different approaches or solutions you considered. ▪ **Decision:** Explain the final decision you made. ▪ **Evaluation:** Share your assessment of the outcome and any lessons you learned.
STAR	▪ **Situation:** Describe the context. ▪ **Task:** Explain your responsibility in the situation. ▪ **Action:** Walk through your approach to solving the challenge. ▪ **Result:** Share the outcomes or results of your actions.

Source- https://www.aha.io/roadmapping/guide/templates/how-to-prepare-for-a-product-manager-interview

Questions for the Interviewers

Some people say this is the last and most crucial stage of an interview. You should prepare questions to ask your interviewers about leadership, processes, and how the product development team work. You should show your interests and enthusiasm and eagerness to learn more about the role and the company values in this stage. This not only helps you assess if the company is the right fit for you but also leaves a lasting impression on the interviewer.

Case Interviews

Practice a product improvement or root cause analysis case with your partner by taking on the role of the interviewee while your partner acts as the interviewer. This exercise will help you develop a structured approach to problem-solving, refine your analytical thinking, and build confidence in tackling case questions. By consistently practicing multiple cases, you will improve your ability to articulate solutions clearly, think critically under pressure, and enhance your overall performance in your final interview.

Some More Tips & Tricks!

Even though having a technical background has advantages to land PM job roles, it is not mandatory and it would require a lot more effort to convert the interview. Skills required to convert the role can be easily learned and honed by candidates from non-technical background as well.

Participating in case competitions, either in B school competitions or corporate ones having specifications to PM helps a lot in skill development and resume building. Sometimes, winning these company competitions can directly award you a PPI.

BEST BOOKS & RESOURCES FOR PM PREPARATION

1. Cracking the PM Interview by Gayle

2. Decode and conquer by Lewis C. Lin

3. Case Interviews Cracked by Kelshikar and Garg

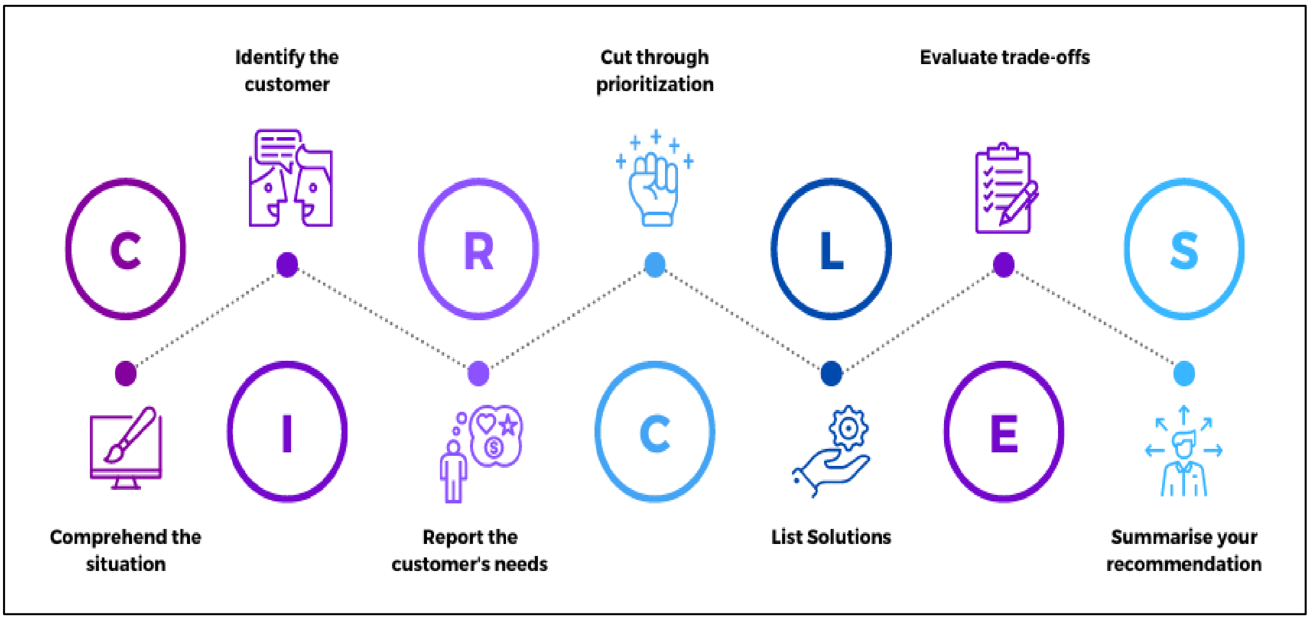
4. The Product Folks YouTube/website

Product Design Frameworks

PRODUCT DESIGN FRAMEWORKS

CIRCLES Framework

The CIRCLES framework is a structured approach to solving product design problems. It helps product managers methodically analyse a situation, identify customer needs, and prioritize solutions effectively. Below is a detailed breakdown of each component of the CIRCLES framework.



Source: Product Stash

C - Comprehend the Situation

Understanding the broader context of a problem is the first and most critical step. A well-defined problem leads to better solutions. In high-pressure environments, truly comprehending the situation is easier said than done.

Key Considerations:

- <u>Clarify the Goal:</u> Identify the objective behind the product update. Is it aimed at increasing revenue, improving engagement, or enhancing user retention?
- <u>Understand Constraints:</u> Recognize limitations such as budget, technology, regulatory compliance, and organizational priorities.
- <u>Analyse the Context:</u> Understand the users, their pain points, & the existing market landscape.

I - Identify the Customer

Defining the target audience ensures that the product aligns with user expectations and requirements. Try to create a persona

Key Considerations:

- Demographics: Age, location, income level, and profession.
- Behavioural Patterns: Usage habits, preferences, and interaction with similar products.
- Pain Points: Challenges faced by users that the product aims to solve.

Persona: Shubham, the busy IT professional

Persona Name	Behaviour	Needs and Goals	Pain Points
Shubham, the busy IT professional	Works 12+ hours a day	To Stay productive, manage tasks efficiently, reduce stress	No time for manual organization, gets overwhelmed with tasks

R - Report Customer Needs

Identifying the target audience's specific needs is a crucial step in gathering and outlining the key requirements and expectations of users. This process ensures that product development stays aligned with their actual demands. Developing use cases tailored to each user persona helps guarantee that.

C - Cut (via Prioritization)

During an interview the time is limited, so you cannot fix every issue you have identified. The key is prioritization—focusing on one or two high-impact customer needs. You might choose based on how severe and frequent the problem is or how much value it creates versus the effort (Value vs Effort, RICE Framework) required. Just like in real-world product development, you will always have a long list of potential improvements, but with limited resources you need always first tackle the most impactful problem first.

L - List Solutions

This is an exciting part to answer you get a chance to showcase your knowledge for product management. A single customer pain point can be addressed in multiple ways. For example, if the issue is long wait times in food delivery, solutions could include an optimized order batching system or a feature that lets users choose priority delivery for an extra fee. If I have identified more than one pain point, I'll outline solutions for each, limiting them to four well-thought-out features that directly address customer needs. Employers are evaluating creativity and product vision, looking for product managers who can anticipate future trends in technology and customer behaviour. They expect you to identify opportunities and execute a strategy that leverages these trends for the company's success.

E - Evaluate Trade-offs

Every decision has pros and cons. Evaluating trade-offs ensures balanced and strategic decision-making.

Key Considerations:

- <u>Cost vs. Benefit:</u> Weigh financial investment against potential returns.
- <u>Complexity vs. Usability:</u> Ensure added functionalities don't compromise user experience.
- <u>Short-Term vs. Long-Term Impact:</u> Consider immediate gains vs. long-term sustainability.

S - Summarize Recommendations

The final step involves 30-60 seconds to summarise you answer

- Which feature do I recommend?
- A quick recap – What the feature is and how it benefits users and/or the company.
- Why this solution? – Reasons for choosing this over other options

By following the CIRCLES framework, product managers can systematically approach problem-solving, ensuring that solutions are user-centric, feasible, and aligned with business goals.

Practise Question 1: Design a Smart Grocery Shopping App

Imagine you are in a product management interview, and you get this question: **"You are a product manager at a tech company, and you need to design a smart grocery shopping app. How would you approach this?"**

Comprehend the Situation (C) – Understanding the Problem

Questions to Ask:

- What does "smart" mean in this context? Is it AI-driven recommendations, automated shopping lists, price comparisons, or something else?
- Who is the target audience? Busy professionals, families, elderly people, students?
- What is the business goal? Drive revenue through subscriptions, increase user engagement, or integrate with grocery stores?
- Are there any constraints? Budget, timeline, technology limitations, partnerships with supermarkets?

Example Question: "Are we building a grocery app with AI-powered personalized recommendations, or are we focusing on price comparison and deals?"

Identify the Customer (I) – Who Will Use It?

Possible customer segments for a smart grocery app:

1. <u>Busy professionals:</u> Want quick grocery delivery and easy meal planning.

2. Families: Need bulk purchases, budgeting, and meal-planning tools.
3. Students: Prefer budget-friendly options, discounts, and recipes.
4. Elderly People: Need simple UI, voice-based search, and home delivery.

Example Decision: "Our primary target users will be busy professionals and families who need fast, convenient, and personalized grocery shopping experiences."

Report the Customer's Needs (R) – What Problems Do They Face?

For busy professionals, main pain points:

- Forgetting to buy essentials.
- No time to compare prices across stores.
- Meal planning is difficult.

For families, main pain points:

- Budgeting and tracking grocery expenses.
- Managing shopping lists for multiple family members.
- Ensuring groceries do not expire before use.

Example: "Busy professionals struggle with meal planning and forget to restock essentials, while families need better budgeting and list management."

Cut Through Prioritization (C) – What Features Matter the Most?

Using the MoSCoW Method:

- Must-have: Smart shopping list, AI-powered personalized recommendations, price comparison.
- Should-have: Integration with grocery delivery services, nutrition tracking.
- Could-have: Voice assistant for adding items, AR-based product scanning.
- Won't-have (for now): Grocery-sharing with neighbours.

Example Decision: "We'll focus on smart shopping lists and AI-powered recommendations first, as they directly solve users' problems."

List Solutions (L) – Brainstorm Ideas

For smart shopping lists:

- AI auto-generates a list based on past purchases.

- The app sends reminders when essentials are running low.
- Users can scan receipts to track what they bought.

For personalized recommendations:

- AI suggests items based on dietary preferences.
- Smart meal planning feature generates a grocery list based on selected recipes.
- Users can compare prices across multiple stores.

Example: "The app will use AI to auto-generate shopping lists based on past purchases and suggest meal plans with one-click grocery additions."

Evaluate Trade-offs (E) – Pros & Cons of Solutions

Each solution has its trade-offs in terms of complexity, cost, and feasibility.

Feature	Pros	Cons
AI-based smart list	Saves time, personalized experience	Requires machine learning, costly to build
Price comparison	Helps users save money	Needs partnerships with stores
Nutrition tracking	Appeals to health-conscious users	Requires manual data input, reducing adoption

Example Decision: "We'll start with AI-powered shopping lists because they offer high impact with feasible implementation."

Summarize Your Recommendation (S) – Final Pitch

"To address the needs of busy professionals and families, we will build a smart grocery shopping app with AI-powered shopping lists and personalized meal-planning suggestions. The app will track past purchases, predict when users need to restock, and generate a shopping list automatically. Future iterations can include price comparison, delivery service integration, and nutrition tracking to enhance the experience."

Practise Question 2: Improving Google Maps for Pedestrians using CIRCLES Framework

Imagine you are in a product management interview, and you get this question:
"How would you improve Google Maps for pedestrians?"

Comprehend the Situation (C) – Understanding the Problem

Questions to Ask:

- What specific problems do pedestrians face when using Google Maps?
- Are we focusing on urban walkers, tourists, or rural users?
- What is the business impact of improving pedestrian navigation?
- Are there technical constraints? (e.g., GPS accuracy, AR capabilities, data availability)

Example Question: "Are we improving pedestrian navigation for daily commuters or tourists exploring new areas?"

Identify the Customer (I) – Who Will Benefit?

User Segments:

- Daily commuters – Need faster, optimized walking routes with real-time crowd data.
- Tourists – Need landmarks, scenic routes, and better offline navigation.
- Elderly or disabled pedestrians – Need accessible routes with fewer stairs and safe crossings.

Example Decision: "We will focus on tourists and daily commuters since they use pedestrian navigation the most and face clear problems."

Report the Customer's Needs (R) – What Problems Do They Face?

Identify frustrations pedestrians experience with Google Maps.

For tourists:

- Difficulty finding scenic routes instead of just the shortest path.
- Limited offline access to pedestrian navigation in new cities.

For daily commuters:

- Traffic lights & crossings not factored into estimated time.
- No real-time crowd density, leading to congested sidewalks.

Example Insight: "Tourists need scenic and offline-friendly navigation, while commuters need real-time crowd and crossing data."

Cut Through Prioritization (C) – What Improvements Matter Most?

We will prioritize based on impact and feasibility.

Using the MoSCoW Method:

- Must-have: Offline pedestrian maps, real-time crowd data.
- Should-have: AI-based scenic route suggestions.
- Could-have: AR-powered navigation for immersive guidance.
- Won't-have (for now): 3D street-level pedestrian simulation.

Example Prioritization: "We will focus on offline pedestrian maps and real-time crowd data first."

List Solutions (L) – Brainstorm Ideas

Generate solutions to improve pedestrian navigation.

For tourists' scenic routes:

- AI-based scenic route recommendations based on landmarks and parks.
- Community-rated walking trails integrated into Google Maps.

For commuters' real-time navigation:

- Traffic light & crossing time prediction to adjust walking estimates.
- Crowd density visualization for better route decisions.

Example: "We will integrate scenic route suggestions for tourists and real-time pedestrian congestion tracking for commuters."

Evaluate Trade-offs (E) – Pros & Cons of Solutions

Each idea comes with its own set of trade-offs in terms of feasibility, cost, and overall impact. While certain ideas may be highly effective in achieving the desired outcome, they might also demand a substantial financial investment or encounter significant challenges during implementation. On the other hand, some alternatives may be easier to execute and more budget-friendly, making them appealing in the short term, yet they might not deliver substantial long-term benefits or sustainable results.

Therefore, conducting a thorough evaluation of each option by carefully analysing these trade-offs is essential for making well-informed and strategic decisions that align with long-term objectives.

Feature	Pros	Cons
Offline Pedestrian Maps	Helps tourists	Requires more storage and data compression
Scenic Route Suggestions	Enhances user experience	Needs AI and user input
Traffic Light Prediction	More accurate ETAs	Requires real-time city data
Crowd Density Visualization	Helps commuters	Needs large-scale user data collection

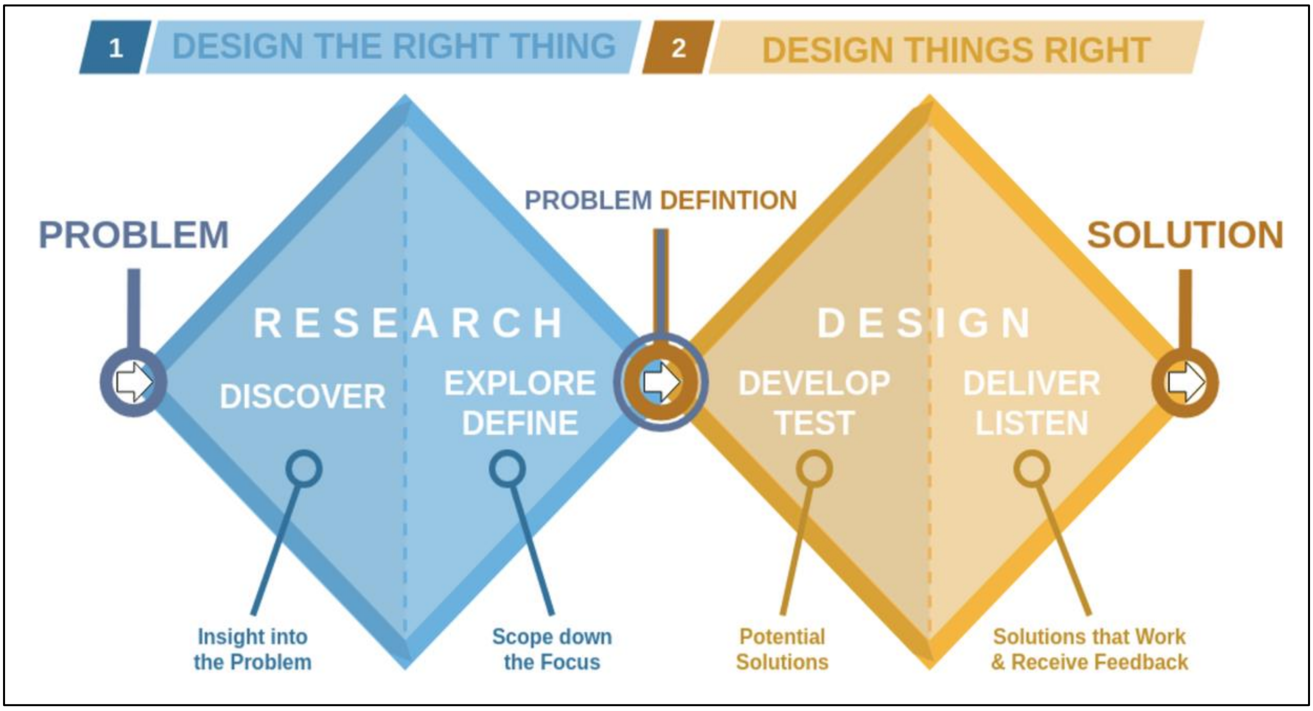
Example Decision: "The offline maps are easy to implement and benefit all tourists, while real-time crowd density tracking improves navigation for commuters."

Summarize Your Recommendation (S) – Final Pitch

"To enhance Google Maps for pedestrians, we will introduce offline pedestrian maps for tourists, allowing them to navigate cities without internet access. For daily commuters, we will implement real-time crowd density tracking to help avoid congested walkways and improve estimated walking times."

Double Diamond Framework

The Double Diamond Framework is a structured problem-solving approach used in design thinking, product development, and policy-making. It was introduced by the UK Design Council in 2005 and helps teams approach complex challenges systematically.



Source: Product Management Made Easy on LinkedIn

The framework consists of two main phases, visually represented as two diamonds:

1. Understanding the Problem (First Diamond)
2. Developing and Implementing the Solution (Second Diamond)

Each diamond is divided into two key steps, making a total of four stages. The framework balances divergent thinking (expanding possibilities) and convergent thinking (narrowing down to focus) at each step.

The Double Diamond Framework ensures that:

- The real problem is explored before defining a solution.
- Multiple ideas are considered before deciding on the best approach.
- Solutions are tested and refined based on feedback.

- The process remains structured, reducing inefficiencies.

First Diamond: Understanding the Problem

The first diamond focuses on problem identification and refinement. It consists of two steps:

Discover (Divergent Thinking – Expanding the Understanding of the Problem)

This is the research phase where information is gathered to get a deep understanding of the problem. Various research techniques are used, such as:

- User Research: Conducting interviews, surveys, and focus groups to understand the needs, pain points, and behaviours of stakeholders.
- Observational Studies: Watching users interact with a product, service, or system in real-life scenarios.
- Data Analysis: Reviewing existing data, reports, or case studies to identify trends and patterns.
- Competitive Benchmarking: Analysing how similar problems have been addressed by competitors or other industries.

At this stage, no conclusions are drawn, and no solutions are proposed.

Define (Convergent Thinking – Narrowing Down the Problem Statement)

After gathering a large amount of information, the next step is to filter and analyse the data to identify the core issue that needs to be solved. This involves:

- Identifying common themes and patterns in research findings.
- Grouping user needs and pain points into categories.
- Prioritizing issues based on impact, feasibility, and urgency.
- Creating a clear problem statement that defines what the problem is, whom it affects, and why it needs to be solved.

A well-defined problem statement prevents unnecessary efforts on symptoms rather than root causes.

Second Diamond: Developing and Implementing the Solution

The second diamond focuses on solution exploration, testing, and implementation. It also consists of two steps:

Develop (Divergent Thinking – Exploring Possible Solutions)

Once the problem is clearly defined, brainstorming and ideation begin. This phase involves:

- Generating multiple possible solutions without filtering them initially.

- Encouraging creativity through structured brainstorming techniques like mind mapping, SCAMPER (Substitute, Combine, Adapt, Modify, put to another use, Eliminate, Reverse), or role-playing.
- Creating rough sketches, wireframes, or low-fidelity prototypes to visualize ideas.
- Gathering feedback from stakeholders and potential users to refine ideas.

The goal here is to think broadly and consider multiple perspectives before choosing a final approach.

Deliver (Convergent Thinking – Implementing the Best Solution)

The final step involves selecting the most effective solution and executing it. This includes:

- Testing the proposed solution with users in small-scale trials or pilot programs.
- Collecting feedback and making necessary refinements based on real-world performance.
- Scaling up and fully implementing the solution while monitoring its impact.
- Measuring success using predefined key performance indicators (KPIs) such as user adoption, satisfaction, efficiency gains, or business impact.

The Deliver phase does not always mark the end of the process. Continuous improvements may be necessary as new insights emerge or as user needs evolve.

Practise Question 1: Design a Mobile Application for Personal Financial Management

Let us say you arere a **product manager** at a fintech startup, and your goal is to design a **personal finance management app** that helps users **track expenses and save money efficiently**.

1st Diamond: Problem Space (Understanding & Defining the Problem)

Step 1: Discover – Exploring the Problem (Divergent Thinking)

You start by researching user behaviours and pain points related to managing personal finances.

- <u>User Interviews & Surveys:</u> You talk to potential users (e.g., working professionals, students, and freelancers) to understand their challenges in budgeting and saving.
- <u>Market Research:</u> You analyse existing apps to see what's missing or what frustrates users.
- <u>Observational Studies</u>: You track how users currently manage their money—spreadsheets, cash, or banking apps.

Findings:

- Many users struggle with tracking small expenses and sticking to a savings plan.
- Users want a simple, automated way to categorize spending without manual input.

Step 2: Define – Narrowing Down the Problem (Convergent Thinking)

From research insights, you frame the core problem:

Problem Statement:

"Users struggle with tracking their expenses efficiently, leading to poor financial planning. They need an intuitive, automated way to monitor their spending and save without much effort."

You prioritize key pain points:

1. Users want automated expense tracking.

2. Users need clear spending insights to improve financial habits.

3. Users prefer a seamless, low-effort experience (no manual data entry).

2nd Diamond: Solution Space (Ideating & Delivering the Solution)

Step 3: Develop – Generating & Testing Solutions (Divergent Thinking)

Now that the problem is well-defined, you brainstorm possible solutions:

- AI-powered categorization of transactions and Smart spending limits with notifications.
- Auto-saving feature that rounds up purchases and transfers spare change into savings.
- Gamification features to motivate users to save (e.g., streaks, rewards).

Build low-fidelity prototypes (wireframes, clickable mock-ups) & conduct usability tests with users.

Step 4: Deliver – Implementing the Best Solution (Convergent Thinking)

Based on user feedback, you finalize the best solution:

- The app automatically syncs with bank accounts to track spending.
- It categorizes expenses using machine learning.
- It offers spending insights & savings recommendations.

You develop an MVP (Minimum Viable Product) and launch a beta version for real-world testing.

Outcome:

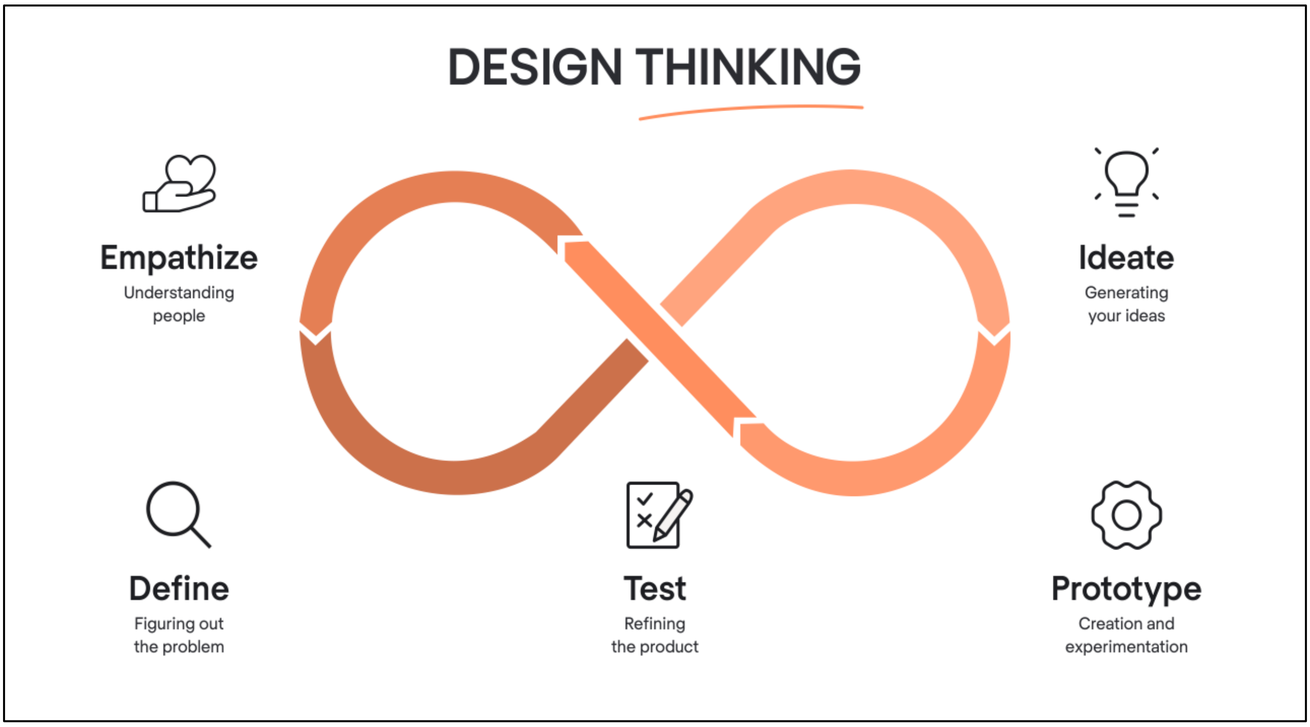
The product goes through iterations, improving accuracy, user experience, and engagement. After launch, you continue measuring user feedback and behaviour to enhance features.

Source: https://productstrategy.co/product-design-frameworks/

Design Thinking

DESIGN THINKING

Design Thinking is a step-by-step process that helps create better products, services, and solutions by focusing on real human needs. Instead of jumping straight into solutions, this approach encourages understanding users first, generating multiple ideas, testing them, and refining them.



Source: Sakshi Gupta on LinkedIn

The 5 Stages of Design Thinking

Empathize – Understand the People You are Designing For

Before creating a solution, you need to understand who you are designing for and what challenges they face. You do this by:

- Talking to users – Ask them about their experiences, problems, and frustrations.
- Observing users – Watch how they interact with products or services in real life.
- Creating empathy maps – Organize what you've learned about their thoughts, feelings, and behaviours.

Goal: Develop a deep understanding of the user's needs and problems.

Define – Clearly State the Problem

Once you've gathered enough information, the next step is to define the core problem. Instead of vague statements like *"People struggle with online payments"*, you create a specific problem statement:

"Busy professionals need a fast and secure way to check their bank balance on the go without logging into a complex banking app."

To define the problem effectively, you:

- Summarize key insights from user research.
- Create user personas – fictional profiles representing different types of users.
- Write a clear problem statement that focuses on what the user truly needs.

Goal: Frame the problem in a way that leads to meaningful solutions.

Ideate – Brainstorm Different Solutions

Now that the problem is clear, the next step is to come up with many possible solutions—even wild or unconventional ones. To do this, you can:

- Brainstorm ideas without judging them.
- Use "Worst Possible Idea" – think of terrible ideas first to spark creativity.
- Look at similar problems in other industries for inspiration.
- Sketch quick concepts to visualize ideas.

For example, if the problem is making online banking faster, some ideas could be:

- A voice-activated bank balance check.
- A one-click balance check without logging in.
- A smartwatch notification that shows the balance automatically.

Goal: Generate a variety of possible solutions before picking the best one.

Prototype – Create a Simple Version of Your Solution

Instead of building a full product right away, you create a basic version (called a prototype) to test your idea. This could be:

- A paper sketch of an app layout.
- A mock-up made in a design tool like Figma.
- A physical model (if designing a product).
- A manual simulation (where someone acts as the system to test the idea).

For the online banking example, you might create a simple app mock-up that shows the one-click balance feature.

Goal: Build something quickly to see if the idea works.

Test – Get Feedback and Improve

Finally, you test the prototype with real users. You:

- Ask them to try it out.
- Observe where they struggle or get confused.
- Gather feedback and note what can be improved.
- Refine the design based on what you learned.

If users find the one-click bank balance feature confusing, you might adjust the button placement or simplify the design.

Goal: Improve the solution based on real user feedback and repeat the process until it works well.

Practise Question 1 – Design Thinking for a Food Delivery App

Imagine a food delivery startup that is facing frequent complaints from users about delayed deliveries, incorrect orders, and a confusing interface.

Empathize – Understanding User Problems

The first step is to understand the challenges users face by gathering insights from customers, delivery partners, and restaurant owners.

- Customer Feedback: Users report delays, missing items, and difficulty navigating the app.
- Delivery Partner Interviews: Riders mention unclear addresses, long wait times at restaurants, and difficulty contacting customers.
- Restaurant Owners' Input: Restaurants state that they receive orders late and struggle to manage peak hours efficiently.

Insights reveal that delivery issues stem from poor communication, unclear addresses, and delays at restaurants.

Define – Identifying the Core Problem

The team synthesizes these insights into a clear problem statement:

"Food delivery customers need a faster and more reliable way to receive their orders, while delivery partners and restaurants need better communication and order management tools."

This ensures the focus remains on the needs of all key stakeholders.

Ideate – Brainstorming Possible Solutions

The team explores various solutions to address the identified problems:

- Live Order Tracking: Provides customers with real-time updates on their order status.
- Optimized Route Mapping: Uses AI-based navigation to help delivery partners find the fastest routes.
- Smart Address Suggestions: Automatically suggests corrected addresses based on past orders.
- Direct Messaging Feature: Enables seamless communication between customers, delivery partners, and restaurants.
- Predictive Order Preparation: Uses AI to forecast peak demand, allowing restaurants to prepare popular dishes in advance.

Prototype – Creating Simple Versions to Test

Instead of making major changes at once, the company develops small-scale prototypes for testing:

- A beta version of live tracking for a selected group of users.
- A prototype of the AI route planner to compare delivery times.
- A chatbot to handle order-related queries and improve customer support.

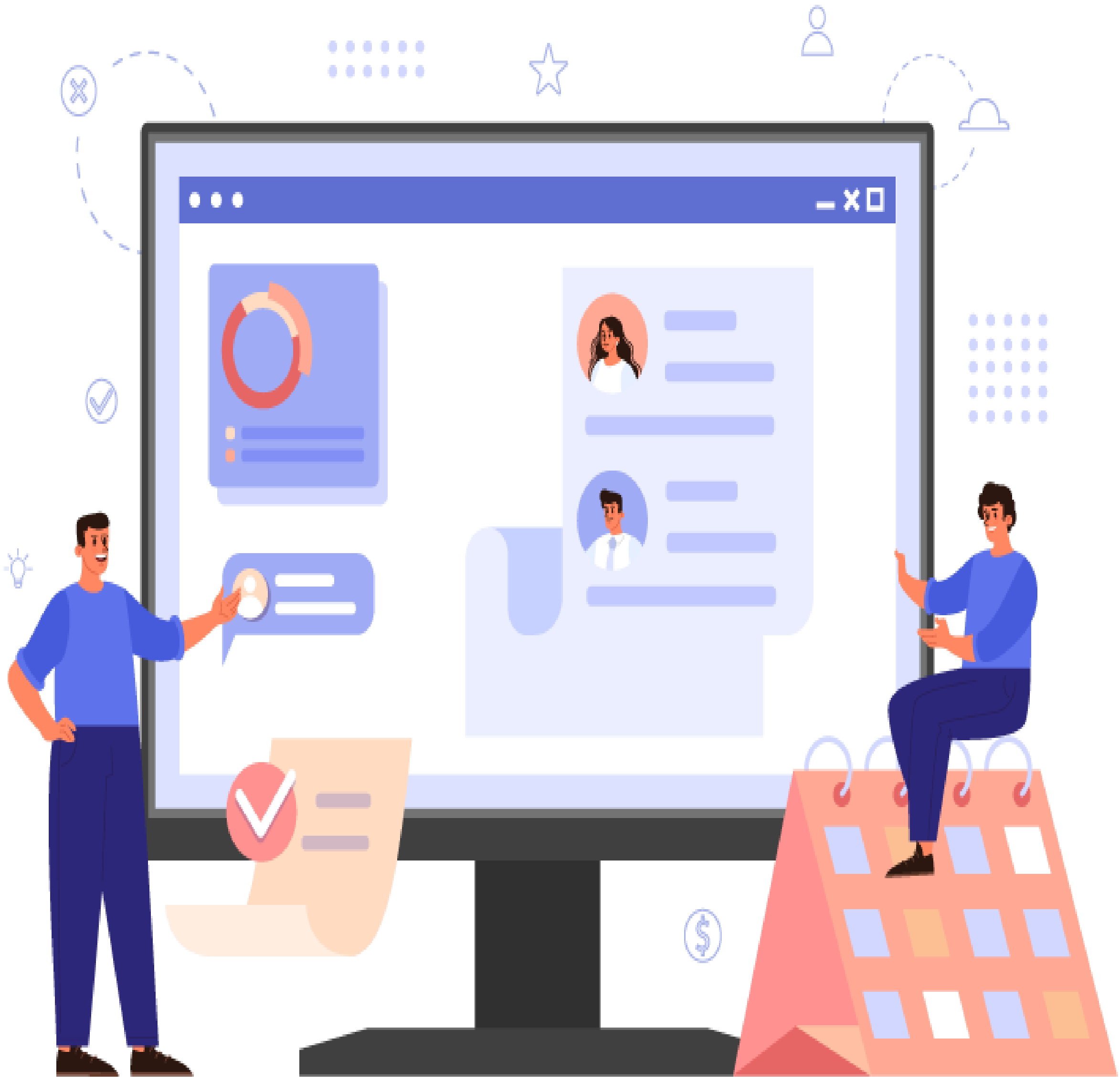
Test – Gathering User Feedback

Users test the new features, and the results show:

- Live tracking reduces customer complaints as they now know when to expect their orders.
- AI route mapping improves delivery times by 15 percent.
- Direct messaging between customers and delivery partners helps resolve address issues more efficiently.
- Some users find chatbot responses limited, indicating the need for further refinement.

Based on this feedback, the chatbot is improved, and AI-based route mapping and live tracking are implemented across the platform.

Source: https://productstrategy.co/product-design-frameworks/



Prioritization Frameworks

PRIORITIZATION FRAMEWORKS

RICE Scoring Method

Prioritizing tasks, features, or projects can be one of the biggest challenges when building a product roadmap. You may have a long list of great ideas, but deciding which ones to work on first is crucial. This is where the RICE scoring model comes in—it provides a structured way to evaluate and prioritize projects based on four key factors:

- **Reach** – How many people will this project impact?
- **Impact** – How much of a difference will it make for each person?
- **Confidence** – How certain are we about the estimates for reach and impact?
- **Effort** – How much work will it take to complete?

Source: GeeksforGeeks

The Four Factors of RICE

Reach – How Many People Will This Affect?

The reach score estimates how many users will interact with or benefit from the project in a given period (e.g., per month or per quarter). This prevents teams from prioritizing features just because they find them personally interesting.

Example: If a feature will be used by 1,000 customers per month for three months,

The Reach score is 3,000 (1,000 × 3)

Impact – How Big is the Effect?

Not all projects have the same level of impact. Some changes make a huge difference, while others are just small optimizations. Since measuring impact exactly is difficult, we use a simple rating scale:

3 = Massive Impact

2 = High Impact

1 = Medium Impact

0.5 = Low Impact

0.25 Minimal Impact

Example: A feature that doubles conversion rates might have an Impact score of 3.

A minor UI improvement might have an Impact score of 0.5.

Confidence – How Sure Are We About Our Estimates?

Some projects have strong data backing them, while others are based on assumptions. The confidence score accounts for this uncertainty.

100% = High confidence (We have strong data to support this.)

80% = Medium confidence (Some data, but some assumptions too.)

50% = Low confidence (Mostly assumptions, little data.)

Example: If we have survey data, product metrics, and engineering estimates, Confidence = 100%.

If we only have a general idea with no data to support it, Confidence = 50%.

Effort – How Much Work is Required?

The effort score measures the total amount of work required across all teams (engineering, design, product, etc.). It is measured in person-months (the work one person can do in a month).

Example: A small tweak that takes one week of work = 0.5 person-months.

A major redesign that requires multiple teams working for four months = 4 person-months.

Practise Question 1 – Feature Prioritization for a Tele-Medicine Startup

You are a product manager at a telemedicine startup that connects patients with doctors via an app. Your team has come up with three potential features, and you need to decide which one to prioritize.

RICE Prioritization Table for Healthcare Tech Features:

Feature	Reach (Users per Month)	Impact (Scale: 0.25 - 3)	Confidence (50% - 100%)	Effort (Person-Months)	RICE Score
AI-Based Symptom Checker	50,000	2 (High)	80% (Medium)	4	**20,000**
Automated Medication Reminder System	70,000	1 (Medium)	90% (High)	3	**21,000**
Virtual Doctor Consultation Booking	40,000	3 (Massive)	70% (Medium)	5	**16,800**

Insights & Prioritization:

1. Automated Medication Reminder System (21,000) – Should be the top priority as it has the highest RICE score, indicating broad reach and relatively low effort.
2. AI-Based Symptom Checker (20,000) – Also a strong feature due to its high impact and moderate effort.
3. Virtual Doctor Consultation Booking (16,800) – While impactful, it requires the most effort, making it a lower priority.

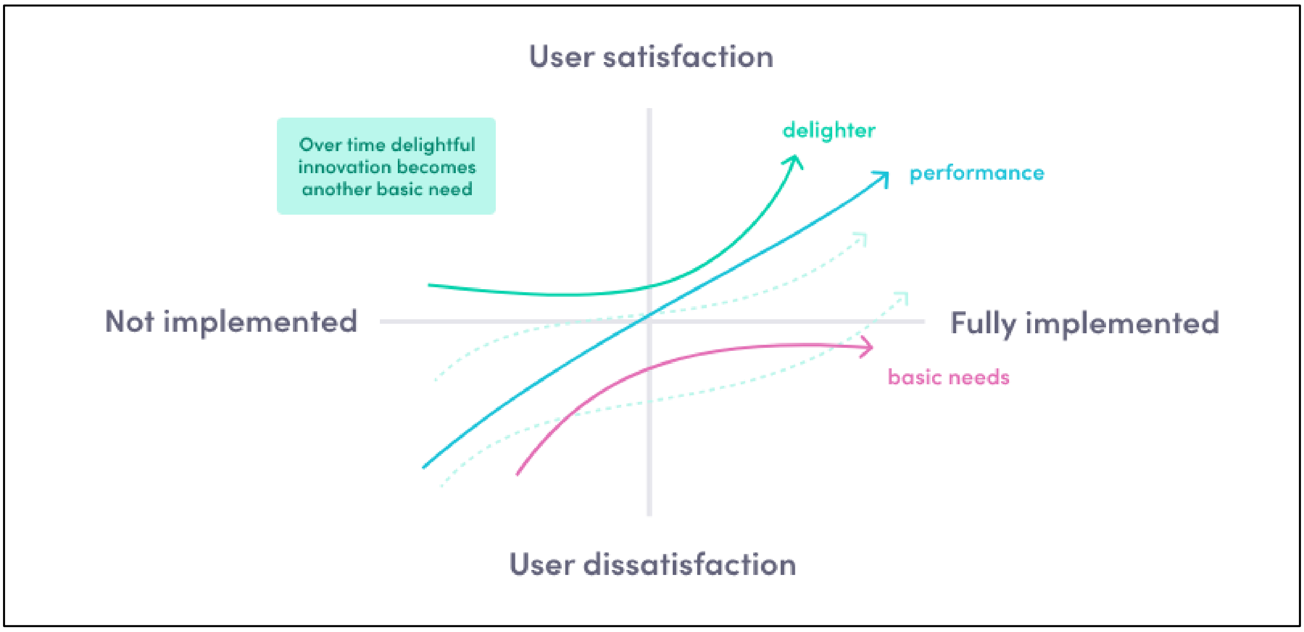
Source: https://productschool.com/blog/product-fundamentals/ultimate-guide-product-prioritization

Kano Model

The Kano Model, developed by Professor Noriaki Kano in the 1980s, is a framework for prioritizing features based on their impact on customer satisfaction. It helps businesses understand which features are essential, which improve satisfaction, and which can create delight.

Categories in the Kano Model

- **Basic Needs (Must-Haves)** – Customers expect these features. Their absence causes dissatisfaction, but having them doesn't necessarily increase satisfaction.
- **Performance Needs** – The more effort you put into these features, the happier the customer. These have a linear relationship with satisfaction.
- **Delighters (Exciters)** – Unexpected features that impress customers and make them love the product. Their absence does not cause dissatisfaction, but their presence creates a "wow" factor.
- **Indifferent Features** – Customers do not care about these features; they neither increase nor decrease satisfaction.
- **Reverse Features** – Some customers might dislike certain features that others love, making them tricky to implement.



Source: Product School

Example: SaaS Productivity Tool

A SaaS-based project management tool (like Asana, Trello, or Notion) can use the Kano Model to prioritize features based on customer expectations.

Feature	Kano Category	Explanation
Task Creation & Assignment	Basic Need (Must-Have)	Users expect to create and assign tasks. If missing, the product is useless.
Integration with Google Drive & Slack	Performance Need	The more integrations available, the more useful the tool becomes.
AI-Powered Task Suggestions	Delighter (Exciter)	Unexpected but impressive—suggesting tasks based on user behaviours can make work smoother.
Dark Mode	Indifferent	Some users like it, but it doesn't significantly impact satisfaction.
Forced Email Notifications for All Actions	Reverse Feature	Some users may find excessive notifications annoying rather than useful.

Source: https://productschool.com/blog/product-fundamentals/ultimate-guide-product-prioritization

Value VS Effort Matrix

The Value vs. Effort Matrix is a prioritization framework that helps teams decide which features, projects, or initiatives to work on first. It does this by evaluating two key factors:

Value – The potential impact of an initiative on business goals and user needs.

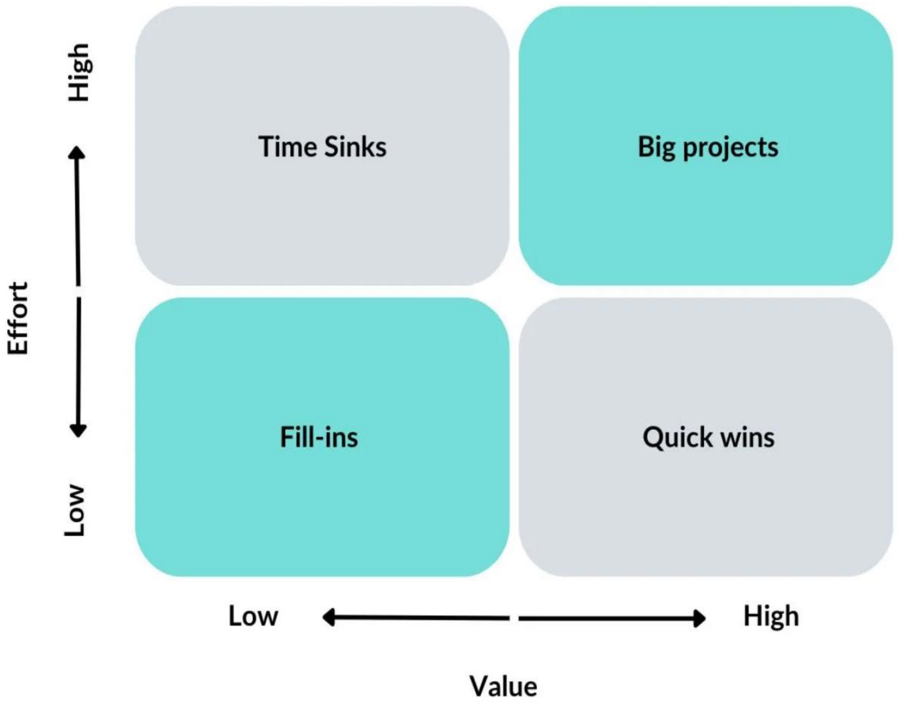
Effort – The resources, time, and complexity required to build or implement it.

How the Value vs. Effort Framework Works

Define Value: What does "value" mean for your business and users? Will this initiative increase revenue, reduce churn, improve retention, or enhance brand awareness? How many users will benefit, and to what extent?

Estimate Effort: How many developer hours, resources, or operational costs will it require? Are there risks involved, such as failure or unexpected complexity? Will it require external services or costly integrations?

Plot on the Matrix: X-Axis (Effort): Low to High and Y-Axis (Value): Low to High



Source: UserXD

Quadrant	Description
Quick Wins (High Value, Low Effort)	Features that bring high impact with minimal effort. These should be top priority.
Big Projects/Bets (High Value, High Effort)	Important but resource-heavy initiatives. Plan these strategically.
Fill-Ins (Low Value, Low Effort)	Minor enhancements that don't take much effort. Can be considered if time permits.
Time Sinks (Low Value, High Effort)	Features that are difficult to implement but don't provide significant value. Avoid these.

Practise Question 1- Applying Value vs. Effort in a SaaS Product Management Tool

Let's say you are a Product Manager for a task management software like Trell or Asanaz. You have multiple feature requests, and you need to prioritize them.

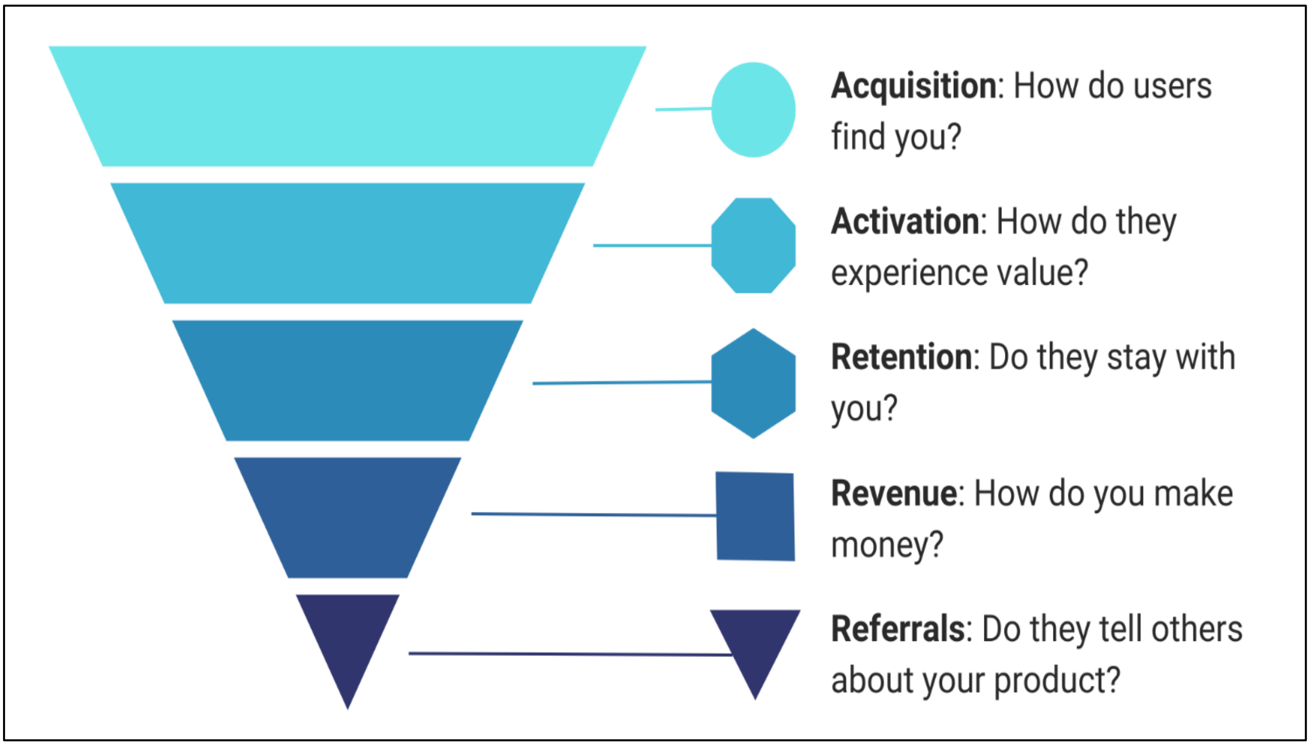
Feature	Value	Effort	Quadrant	Explanation
Task Auto-Suggestions	High	Low	Quick Win	Helps users complete tasks faster with minimal development work.
Advanced Analytics Dashboard	High	High	Big Bet	Valuable for enterprise users but requires significant backend effort.
Customizable UI Themes	Low	Low	Fill-In	Aesthetic improvement, but does not impact core functionality.
AI-Powered Project Planning	Low	High	Time Waster	Complex to build, and most users won’t use it.

Product Metrices

METRICES

Pirate Metrics (AARRR Framework)

The AARRR framework, also known as Pirate Metrics, is a structured model that helps businesses measure and optimize their customer journey. It was introduced by Dave McClure, as a way for companies to focus on data-driven growth rather than subjective decision-making.



Source: The Product Compass

Acquisition is about getting new users to discover your product through various marketing channels. The goal is to bring in high-intent users at a low cost and ensure they take the first step toward becoming customers.

Key Metrics:

- Website Traffic – How many people visit your site?
- Click-Through Rate (CTR) – How many users engage with your ads or content?
- Customer Acquisition Cost (CAC) – How much does it cost to acquire one customer?
- Signup Conversion Rate – What percentage of visitors sign up for the product?

Acquisition Strategies:

- SEO & Content Marketing – Creating high-quality blog posts, guides, and videos to attract organic traffic.
- Paid Advertising – Running targeted Google Ads, Meta Ads, or LinkedIn Ads.
- Referral & Affiliate Programs – Offering incentives to existing users for bringing in new customers.

- Influencer Marketing – Partnering with key opinion leaders to promote the product.

Activation is getting users to sign up is only the beginning. Activation focuses on getting them to experience value quickly—an "aha" moment where they realize why your product is useful.

Key Metrics:

- Activation Rate – Percentage of users who complete a key action after signing up.
- Onboarding Completion Rate – Percentage of users who finish the guided setup.
- Time to First Value – How long it takes for a user to see the product's benefit.
- Free-to-Paid Conversion Rate – Percentage of free users upgrading to a paid plan.

Activation Strategies:

- Streamlined Onboarding – Using tooltips, product tours, or interactive walkthroughs to guide users.
- Personalized Recommendations – Suggesting relevant features based on user preferences.
- Gamification – Rewarding users for completing setup steps (e.g., badges, points).
- Customer Support & Live Chat – Offering real-time assistance for users who struggle during onboarding.

Example: Facebook discovered that users who added seven friends in 10 days were more likely to stay engaged. This became a key activation metric, shaping their early growth strategies.

Retention measures how often users return to engage with your product over time. A high retention rate means your product is valuable, while a low retention rate indicates that users are dropping off after the first interaction.

Key Metrics:

- Daily Active Users (DAU) / Monthly Active Users(MAU)sS – How often users engage with the product.
- Churn Rate – Percentage of users who stop using the product.
- Customer Lifetime Value (CLV) – The total revenue a user generates over their lifetime.
- Engagement Metrics – Time spent on the platform, number of logins, or actions taken per session.

Retention Strategies:

- Email & Push Notifications – Sending reminders, updates, and personalized recommendations.
- Loyalty Programs – Rewarding long-term users with discounts or perks.
- Feature Engagement Tracking – Identifying underused features and educating users on how to leverage them.
- Community Building – Encouraging discussions, webinars, and user-generated content.

Example: Netflix retains users by personalizing content recommendations based on viewing history, making users more likely to return.

Referrals happen when happy users bring in new users through word-of-mouth marketing. If your product provides exceptional value, people will naturally recommend it to their network.

Key Metrics:

- Net Promoter Score (NPS) – Measures how likely users are to recommend the product.
- Number of Referrals per User – How many people each user invites.
- Referral Conversion Rate – Percentage of referred users who sign up.

Referral Strategies:

- Referral Incentives – Offering rewards for referring new users (e.g., Dropbox's free storage for referrals).
- Easy Social Sharing – Allowing users to share content, achievements, or invitations with one click.
- Exclusive Perks – Providing VIP features or discounts for users who refer friends.

Example: Dropbox used a double-sided referral program, giving free storage to both the referrer and the invitee. This droves massive growth.

Revenue measures how effectively your product turns engaged users into paying customers. This is where the business becomes profitable and sustains long-term growth.

Key Metrics:

- Monthly Recurring Revenue (MRR) – Predictable revenue generated per month.
- Average Revenue Per User (ARPU) – The amount of revenue generated per active user.
- Customer Payback Period – How long it takes to recover CAC.
- Upsell & Expansion Revenue – Revenue generated from upgrades and additional purchases.

Revenue Strategies:

- Freemium to Premium Upsell – Offering basic features for free but charging for advanced features.
- Subscription Tiers – Providing different pricing plans based on user needs.
- Cross-Selling & Bundles – Encouraging users to purchase add-ons or complementary products.
- Annual Billing Discounts – Offering discounts for upfront yearly payments to increase cash flow.

Example: Spotify offers a free plan with ads but monetizes users through premium subscriptions with ad-free listening and offline downloads.

Practise Question 1: Applying AARRR to an Online Fashion Store

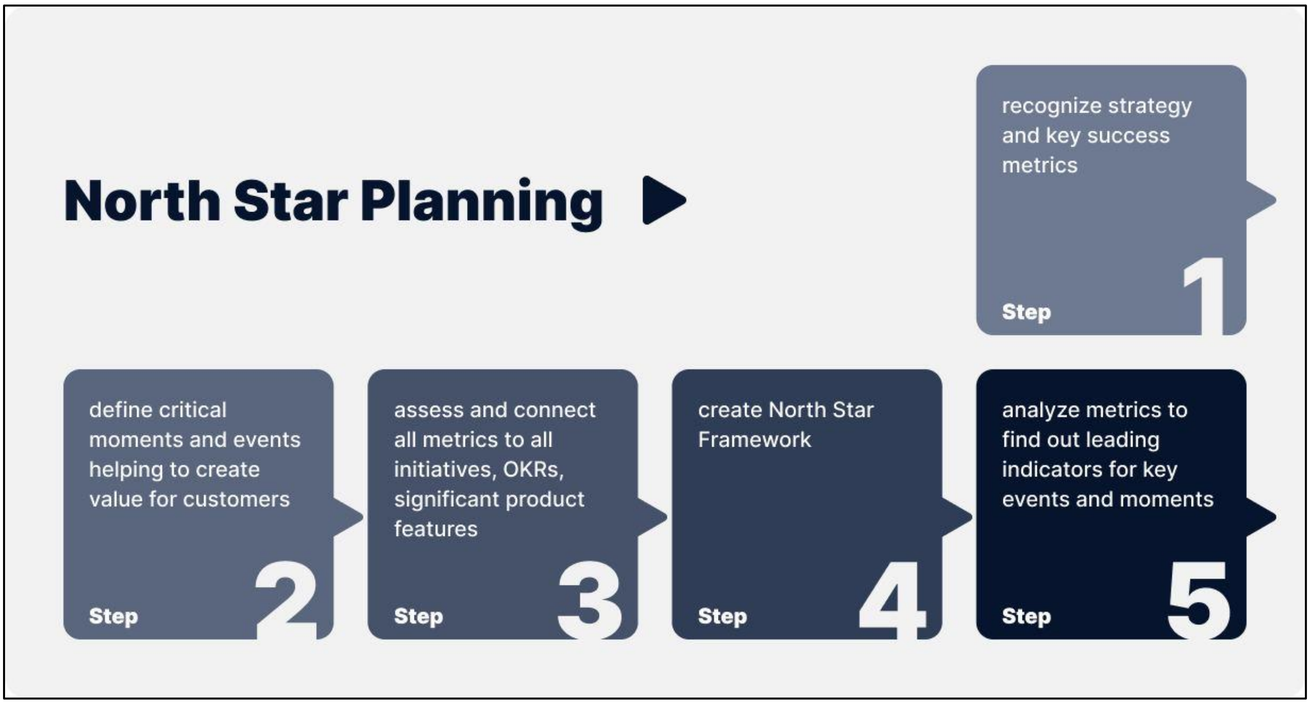
Stage	How It Works in an E-commerce Store	Key Metrics
Acquisition	Customers discover the store through Google Ads, Instagram promotions, influencer partnerships, and SEO-optimized blogs about fashion trends.	- Website traffic - Click-through rate (CTR) - Customer acquisition cost (CAC)
Activation	A customer browses products, adds items to their cart, and completes their first purchase. The store ensures a smooth checkout process with a one-click payment option.	- Cart abandonment rate - First purchase conversion rate - Time to first purchase
Retention	The store sends personalized emails with discounts, offers "back in stock" notifications, and launches a loyalty program to encourage repeat purchases.	- Repeat purchase rate - Email open & click rates - Monthly active shoppers (MAS)
Referral	Customers share referral codes with friends to earn discounts. Social media users post about their purchases using a branded hashtag, helping attract new customers.	- Number of referrals invites - Referral conversion rate - Social shares using the brand hashtag
Revenue	The store increases its revenue through upselling (recommended items), cross-selling (bundling accessories), and offering premium membership for exclusive discounts.	- Average order value (AOV) - Customer lifetime value (CLV) - Subscription revenue from VIP members

Source: https://amplitude.com/blog/pirate-metrics-framework

North Star Metric

A North Star metric is essentially the one measurement that best predicts a company's long-term success. To truly qualify as a North Star metric, it needs to do three things:

- **Lead to Revenue:** The metric should have a direct connection to generating income for the company.
- **Reflect Customer Value:** It must capture the core value that customers receive from your product.
- **Measure Progress:** It should be something you can track over time to gauge the company's growth and overall performance.



Source: Artkai

How to find the right North-Star for your product?

Finding the right North Star metric for your product means identifying the one measurement that is critical to your business's success. In any company, there are many factors at play, but a North Star metric zeroes in on those "load-bearing" pillars—those elements so vital that if they fail, the whole business suffers.

To qualify as a North Star metric, it must satisfy three key conditions:

1. **Lead to Revenue:** The metric should have a clear, direct connection to generating income. If it only measures something that doesn't translate into revenue, it won't drive sustainable growth.

2. **Reflect Customer Value:** It needs to capture the core value your product delivers to your customers. In other words, it should show that your users are truly benefiting from what you offer.

3. **Measure Progress:** It must be trackable over time so that teams can see if their efforts are moving the company in the right direction. This allows for actionable insights and adjustments in strategy.

A North Star metric is not chosen arbitrarily. You can start by asking yourself:

“What is absolutely essential to the functioning of our business?”

“Which KPIs and metrics best capture the most important aspects of our success?”

“What single metric encapsulates customer satisfaction, profitability, and overall progress?”

Once you answer these questions, build a metric hierarchy with your North Star at the top. This process helps ensure that every department is aligned and focused on improving that one measurement.

Sample North-Star metrics as per Industry

Industry	North Star Metric	WHY was this selected
E-commerce	Number of repeat purchases per customer	Focuses on customer loyalty and long-term value by measuring how often customers shop again.
Streaming Services	Total watch time per user per month	Reflects user engagement and satisfaction, driving retention and potential revenue from subscriptions.
Ride-Sharing	Number of rides completed per user per month	Indicates high demand and consistent usage, which correlates directly to revenue and market penetration.
Social media	Daily active users (DAU) engaging with content	Measures active engagement, which is critical for ad revenue and long-term platform growth.
EdTech	Number of lessons completed per user per month	Demonstrates the value provided to users, as consistent learning leads to better retention and success.
Food Delivery	Number of orders completed per active user per month	Higher order frequency means more revenue and increased reliance on the service.
Fitness & Health Apps	Number of workouts tracked per user per month	Tracks user activity and engagement, key for retention and encouraging healthy habits.
FinTech	Total transaction volume per user per month	Indicates trust and consistent usage of financial services, directly impacting revenue generation.
SaaS (B2B)	Number of active teams using the product weekly	Active usage by multiple team members is a strong indicator of product stickiness and renewal potential.
Hospitality & Travel	Number of nights booked per user per year	Reflects customer satisfaction and trust, which drive repeat bookings and overall platform success.

Guesstimates

GUESSTIMATES

Guesstimates are estimates based on logic and calculation. They can be given independently or as a part of a case. The interviewer is generally looking to assess:

- Structure of your approach
- Comfort level with numbers and quick calculations
- Ability to make back of the mind checks to validate the numbers

As a part of the guesstimate, you will be required to make reasonable and logical assumptions, it is important to state them clearly and seek confirmation from the interviewer. You will not be expected to calculate the exact value, but a rough number.

Approaches to Solving Guesstimates

Top-Down approach

Involves starting with an entire population (in other words, the "top" level) and then breaking it down until you arrive at an answer.

The major elements of a top-down approach are:

- Identifying the correct starting universe
- Applying the right set of filters and relevant conditions. Segmentation can be carried out as follows:
    - Demographics (age, sex, income, ethnicity, etc.)
    - Psychographics (attitudes, behavior, values, e.g. smoker vs non-smoker etc.)
    - Geography (urban vs rural, Tier I/II/III cities, country, continent, etc.)
    - Channels (offline vs online, mobile app vs website, etc.)

Bottom-Up Approach

For this approach, rather than starting from the "top" with a high-level figure such as population, the best approach is to start from the "bottom" some low-level statistic, such as Revenue per customer, and build your way up to the answer.

Some common bottom-up approaches:

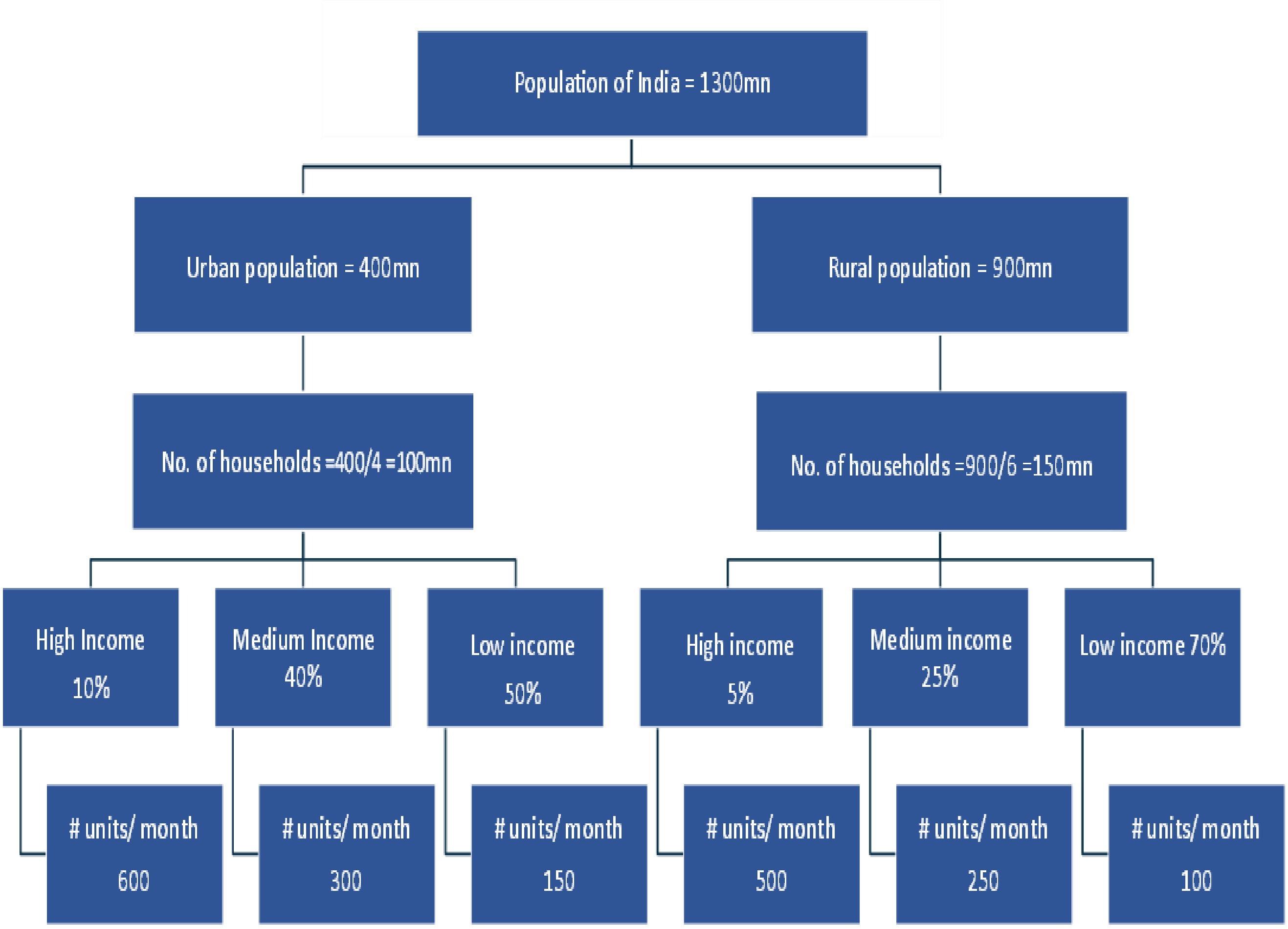
1. **Household Approach**
   Example: For a guesstimate involving the number of cars, washing machines etc. driving calculation using a household approach is easier

2. **Population Approach**
   Example: For estimating the number of mobile phones, chocolates consumed, apparel bought a population approach can be used

3. **Bottle-neck Approach**

Example: For estimating the number of flights per day in a busy airport the number of available runways would serve as a handy bottleneck to base your calculations. There could also be a limitation on the production capacity of the manufacturer in certain cases.

Guesstimate 1: Monthly Residential Electricity Consumption in India

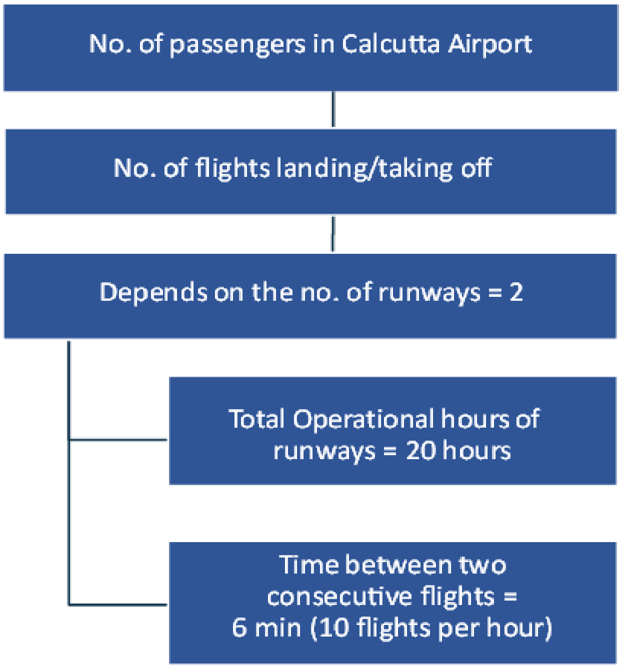


Here # denotes "number of"

Notes and Assumptions:

- The 40/60 split of urban/rural population has been considered
- The average household size assumed for urban and rural areas is 4 and 6 respectively
- Additionally, division of the income groups into high, medium, and low has been considered. The candidate can confirm the same with the interviewer before proceeding

Guesstimate 2: Daily passengers in Calcutta Airport



Hence total flights handled in a day = 2*20*10

= 400 flights

Total domestic passengers per day = 400*70%*180*70%

= ~35,300

Total International passengers per day = 400*30%*400*70%

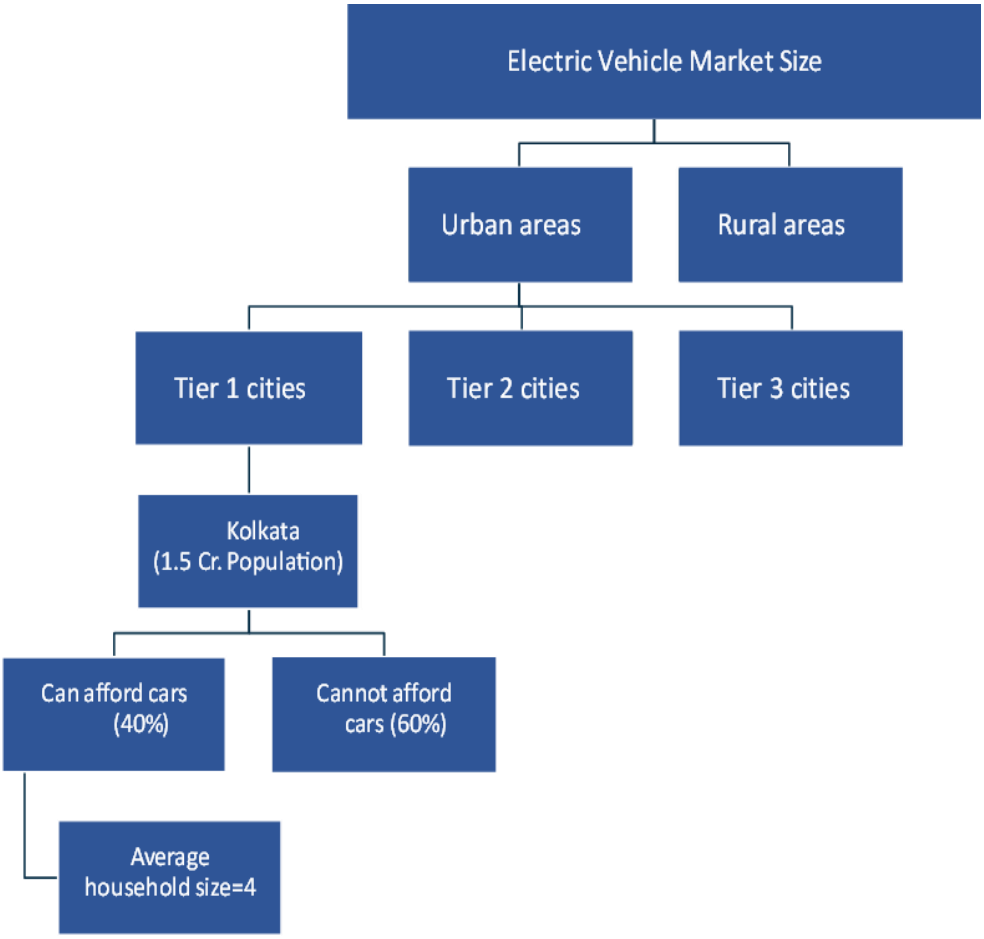
= 33,600

Hence total daily passengers = 69,000(35300+33600)

Notes and assumptions:

- Daily number of passengers depend on the number of flights handled which depends on the number of runways
- The total operational hours of each runway has been considered to be 20 hours (an alternate approach can be considering peak hours for the runway)
- Split of domestic vs international flights is 70/30

Guesstimate 3: Estimate Electrical Car Market Size in India/ Any City



Total number of cars in Kolkata = (1,50,00,000/4) * 0.4

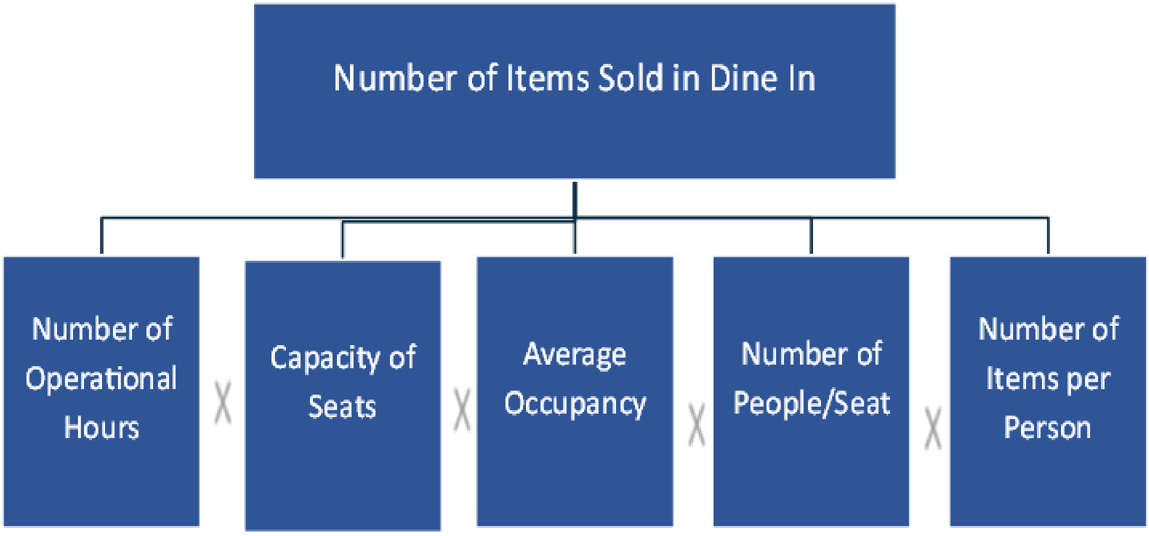
=15,00,000

Assuming a market penetration of 1% = 15,000 electric vehicles

Notes and assumptions:

- The calculations have been performed for a specific example of Kolkata, the same approach can be extrapolated to other Tier 1 cities.
- Rural areas, tier2 and tier3 cities can be neglected stating a 80/20 rule, that majority of the electric vehicle penetration would be in Tier 1 cities.

Guesstimate 4: Total Revenue of a Fast-Food Chain



Number of operational hours = 12 hours

Capacity (Number of Seats) = 50

Average occupancy= 70%

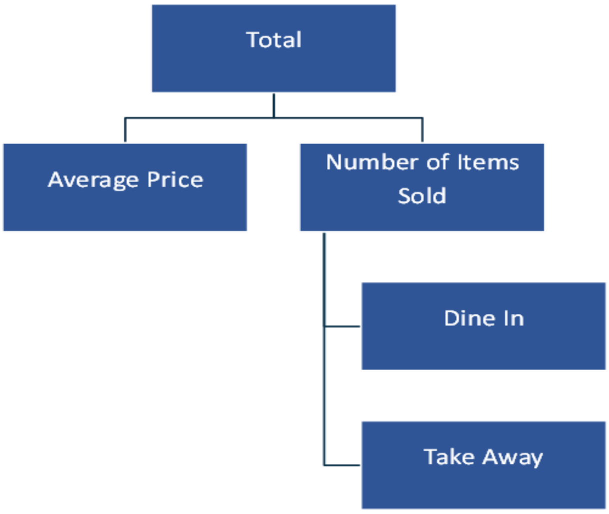
Number of items per person = 2

Number of people/seat/hr = 12 * 50 * 0.7 * 2 * 2

Number of people/seat/hr = 1680 items

Notes and assumptions:

- The total revenue of a fast-food chain is through two sources dine-in and takeaway.
- The seating capacity of the restaurant, along with the number of Mutually Exclusive and Collectively Exhaustive (MECE) factors, is essential for accurately computing the total number of items sold through drive-in services. This calculation can be effectively managed through a comprehensive mathematical formula that considers multiple operational variables.



Guesstimate 5: Market Selection for a Client {Mock Interview}

Client is looking forward to invest in production of Lithium-ion batteries and want our help in selecting the place for setting up its plant.

Interviewer: Hi, Good Morning!! How are you?

Candidate: Hello, I am good. How are you?

Interviewer: So, let's start with the case. We will discuss about the Lithium-ion batteries. Do you have any idea about them?

Candidate: Yes, I know about the Lithium-ion batteries. They are pivotal for EV market and are extensively used in many other industries such as smartphones

Interviewer: Good. So, there are two ways of extracting lithium. One from extracting it from the underwater pools and other from solid rocks. Basis these methods the market for extraction of Lithium is segregated in two ends of the world. One is South America and the other is China. What factors you think are relevant here in this selection

Candidate: This is quite interesting information. I think there are multiple things to look at while making this decision. I would like to ask a few clarificatory questions. For our client who is the key customer. I believe that major usage of batteries are in EV and in that case the market is US because of Tesla setting up battery manufacturing unit and influx of EVs. If it is smartphones oriented, then in that case market should be China and Korea.

Interviewer: That is a good observation. Our client is focusing the Electric Vehicle segment which will have 60% of Lithium demand as per reports.

Candidate: Great so keeping this thing in mind. I think that our client need to compare the cost of setting up the plant at both locations. It may be impacted by the geo-political and economic issues. One important factor to consider is the cost of transport because there is a significant difference in location of two sites from key market (USA).

Interviewer: Well, these are good points. As far as the transport costs are concerned, they can be rationalized and can also be managed because of cost of production. Let me give you some more information on the cost of extraction and production. In China the labour is very cheap but the capital cost is high. On the other hand, in Latin America the labour is not that cheap.

Candidate: Oh, so in that case analysis need to be done. It is a classic case of difference in fixed and variable costs. In one case (China) the fixed cost is high, but the variable cost is low. On the other hand, in case of Latin America variable cost is high and fixed cost is high. I also want to ask here that is there any difference in the quality of the product that is produced in China and Latin America

Interviewer: Well yes you have identified the issue correctly. It is in fact one of the major considerations. Our client has the cost measures of both countries. So how you propose to decide here. Also to answer your question there is no significant quality variation.

Candidate: So, I think battery manufacturing is a mass produce product. Volumes are very important for such a set up. I will identify the breakeven point of volume. After a particular volume size, the difference in fixed costs can be recouped by the benefit occurring on the variable costs. Say if China has 1,00,000 additional fixed cost but Rs 2 benefit in variable costs. Then after a pivot point of 50000 units it will be beneficial to me to set up factory in China. We need to assess how much production/demand is expected and accordingly decide if it is below or above the pivot point.

Interviewer: Okay this is good. Thank you.

Guesstimate 6: Market Entry for a Pharma Firm {Mock Interview}

Interviewer: Hi, good to see you! Without wasting any time let us get to business immediately. Our client is a pharmaceutical major in and is looking to enter the snow Melter liquid chemical market in US. They need us to estimate the market size and how they should enter the market.

Candidate: Sir, to make sure I have understood the context clearly, our client is a pharmaceutical company, and they are considering expansion into snow Melter liquid chemical market in the US. I am required to estimate the markets size, analyse the current market players and decide whether to enter or not enter the market.

Interviewer: That is correct!

Candidate: Sir, if I may inquire any particular reason why they want to enter the market.

Interviewer: They think that there is a large opportunity in the market since the market share is quite fragmented, however they are not sure how big the opportunity size is.

Candidate: Okay. I would like to first like to take a shot at estimating the market size and then think of the business opportunity. I would like a moment please to jolt down.

Interviewer: Sure, go ahead and let me know what information you need.

Candidate: Sir, I would first understand the market area – regions where there is regular snowfall and the product is most likely to be used. Areas with scanty snowfall are less likely to use the product as the weather would self-clear the snow (Demand side estimation).

Then, based on the average snowfall in these locations (thickness * volume) I would estimate the amount of snow that needs to be thawed. Based on the frequency of usage per year and the amount of liquid needed to thaw per unit volume of snow,

I should be able to estimate the volume of chemical needed in the market. Please let me know if this approach is correct.

Area with frequent rainfall ->Amount of snowfall->frequency of usage- >liquid needed

Where Volume of liquid needed = Area * snowfall amt. (height) * frequency (per annum) * Liquid /unit snow

Interviewer: Okay. How do we reach to an approximate area where it snows heavily?

Candidate: Well, I am aware of approximate area of India and I would assume that USA is roughly 3-4 times our country's size. Further I would need a weather report indicating areas with above average rainfall.

Interviewer: So, can you calculate the area for me.

Candidate: Assuming India is a double triangle shaped body,

Area = 2*(1/2*base*height).

Base = East to west

Length = 2500 km

Length = (North *South length)/2 = 1200 km

Area of India = 3,000,000 km2

Area of US = 10, 000,000 km2

Sir, do we have the reports or should I assume a fraction of the area would be our target market?

Interviewer: Assume 70% area is under consideration. The snow is thawed 20 times in a region on an average, average rainfall is 100 cm and per unit (metre cube) snow 10 mL of liquid is required. Can you quickly estimate the volume needed every year?

Candidate: Sure sir.

7000000*1000*1000*1*20*0. 01

=7000 billion *20 * 0.01 litres

= 1400 billion litres

Sir, what is the cost per litre of this liquid.

Interviewer: 6$ per litre.

Candidate: Okay, that makes the overall industry roughly 10$ billion in size.

Interviewer: You sure, the industry size is this big. Check for the zeroes.

Candidate: Yes, sir I think I am right.

Interviewer: So, what is your understanding should our client enter into the market or not?

Candidate: Sir, I would like to understand the current distribution of players. Who are the major players and what is the industry growth rate?

Interviewer: The industry has 8-10 players currently and none of them have a share of more than 8%. There is not much happening in the industry, margins have also saturated.

Candidate: Okay. Interesting! I am trying to understand that given a fragmented industry, our client might want to consolidate its position and emerge as the market leader. However, I am bound to think otherwise given the industry has reached its maturity.

Interviewer: That is correct.

Candidate: Sir, I think I need to take a pause to assess if I have analysed the complete situation.

Interviewer: Sure!

Candidate: Sir, I am trying to think why the company wants to enter the industry. Is the product that they are offering any different?

Interviewer: There you are, so our client has come up with a patented solution in independent research which thaws snow using lesser amount and this would bring down the cost of thawing. So, given the scenario, how should the client enter the market?

Candidate: The client could either enter independently, acquire small sized players in the market, or do a joint venture with one of the major competitors. I would suggest given the inexperience of the client and the industry and the strong R&D support, it would be better if the client goes into a Joint Venture with a couple of the existing local players using their B2B local distribution network and building strong product capabilities through its R&D focus.

Interviewer: Great! That will be all. You can wait outside.

Guesstimate 7: Profitability for a PSU Bank {Mock Interview}

Interviewer: Good morning, take a seat. How has the day been for you so far?

Candidate: Thank you, sir! The day has been good so far. This is my second interview of the day and I think my first interview went well.

Interviewer: Okay. You have been editor to college magazines. Do you also write?

Candidate: Yes, sir I write sometimes however, mostly that is for my own sake. I kind of use it as a medium to give vent to my thoughts.

Interviewer: Good., do you use a debit card?

Candidate: Yes sir. I do have a SBI card. I use it quite often.

Interviewer: And do you have other savings bank accounts apart from SBI?

Candidate: Yes, sir I have an account with PNB; however, I do not use it that often. Little did I realise that the case revolved around the above statement of mine.

Interviewer: Our client is a large PSU Bank and is losing out on profitability. You have been hired to solve the problem for me. They have currently 20 million savings/fixed deposit account.

Candidate: Sir, is this an industry wide phenomenon?

Interviewer: No. Let us just focus on the client only.

Candidate: Sure. So, can I assume that the interest rates and CRR regulations etc. are same for competitors and there have been no major changes?

Interviewer: Yes, absolutely.

Candidate: Sir, since this is about profitability I would like to first look at the revenues and then the costs to understand what could be troubling the company.

Interviewer: Okay.

Candidate: Sir, give me a moment to lay down the revenue and cost drivers for a bank.

1. Revenue: Interest income, commissions, brokerage income
2. Costs: Interest expense, loan defaults, SG&A, Utilities, advertising, employees, loan premium

Sir, these are the areas I would like to inspect to assess the profitability

Interviewer: Okay, let's go ahead with your approach

Candidate: Sir, I am trying to understand if the interest income from our investments/loans has declined or there is a decrease in the brokerage income in the past few years. This might have affected the company's bottom line.

Interviewer: I do not think that has been a problem for us. In fact, our loan portfolio has increased with the launch of the loan advisory portal.

Candidate: Sir, have the overall revenues declined over the past year or it has increased? I should have asked this first before jumping onto the revenue drivers.

Interviewer: I think our top line has been steady. Look, I think the case is not really going in the right direction. Can you rethink about it!

Candidate: Sir, should I investigate the cost drivers? I might find something there.

Interviewer: Go ahead!

Candidate: Has the default ratio increased, have we written off more debt this year than previous years?

Interviewer: Look, I think we have spent a lot of time and we are not really going anywhere. Would like you to take a deep breath and think of an alternative approach. If it helps let me tell you that the total money currently deposited lying in the savings bank account of the client is INR 5000 Cr.

Candidate: Okay. Can I take a minute to pause here.

Comment: Though the interviewer was calm and supportive, I guess I was now under a little pressure (okay maybe I was totally clueless and nervous as to what I should do!) I spotted the no of accounts opened written in the corner of the page and suddenly realised I had not used that information at all.

Candidate: Sir, INR 5000 Cr deposits and 20 million accounts that makes INR 25000 deposit per person.

Interviewer: Are you sure of the figure!

Candidate: Sorry, sir it is 2500 per person. Now that I am looking at this number, I think I completely went wrong on my approach.

Interviewer: okay! Now that you have realized that can you quickly tell me where we went wrong, what would be the new approach and what would be your hypothesis. We don't have much time left.

Candidate: Sure sir, I think I missed out on the customer perspective totally in the beginning and our how our services might have affected our old/new customer base. My alternative approach in that would have been to understand the various customer segments and see if we are losing out on our customer share either the existing ones or the new ones.

Interviewer: Customers is fine but you still have not got it completely. What if I tell you that new customers are daily opening our account?

Candidate: In that case, as I calculated earlier INR 2500 per month means that they are not saving enough money in our bank. To further diagnose the problem, I would like to isolate accounts which have been dormant for more than a year. Some of the possible reasons for this could be:

Poor service as compared to competitors, Low Interest rates, Poor income of a certain set of customers, Alternate investment options for customers

Interviewer: Now we are talking sense. Can you also tell me how dormant accounts could be hurting the bank?

Candidate: Sir, this would happen in three ways:

1. This might hurt the liability side of the balance sheet of the bank. Less deposits would require the bank to look for alternate sources of funds for asset management.
2. Bank's liquidity and solvency would also be affected.
3. The bank would unnecessarily bear the cost of carrying dormant accounts.

Interviewer: Thanks. You can go now.

Root Cause Analysis

ROOT CAUSE ANALYSIS

Understanding Root Cause Analysis Questions

Root cause analysis (RCA) questions are a common type of execution-related inquiry in product management interviews. These questions present a scenario where an unexpected issue has occurred with a product, and as a PM, you must diagnose and resolve the problem.

For example, an interviewer might ask:

"Imagine you are a PM at Rapido, and there has been a 15% increase in ride cancellations. What would you do about this?"

Unlike other PM interview types, RCA interviews follow a "role-play" format, where the interviewer acts as an uninformed data analyst. Your task is to ask relevant questions, analyse available data, and refine hypotheses until you pinpoint the root cause.

Key Skills Interviewers Look For

While arriving at the correct root cause is important, your approach to problem-solving is what truly matters. The following skills will help you earn a "strong hire" recommendation:

- Analysing situations with limited data
- Developing and refining hypotheses iteratively
- Asking data-driven questions to gather insights
- Identifying the root cause using logical reasoning.

Framework for Answering RCA Questions

To effectively answer RCA questions within the typical 30-minute interview timeframe, follow this six-step framework:

Step 1: Clarify and Gather Context

Begin by asking clarifying questions to define the problem's scope and gather relevant context. Ensure you understand key terms and metrics.

Example Questions:

- What counts as a ride cancellation?
- Are cancellations happening before or after a driver is assigned?
- Is this affecting both drivers and riders?
- Is the 15% increase measured week-over-week or month-over-month?

Step 2: Form High-Level Hypotheses

Once you understand the problem, develop broad hypotheses about potential causes. These should cover different aspects of the product and operations:

Possible Hypotheses:

- Technical Issue: A bug in the app might be causing automatic cancellations.
- Product Change: A recent update may have unintentionally encouraged cancellations.
- Operational Change: A change in driver pay or incentives could impact cancellations.
- External Factor: Events like bad weather or a major concert could affect ride behaviour.

Clearly communicate your hypotheses to the interviewer before prioritizing them.

Step 3: Gather Data

Now, refine your hypotheses by gathering relevant data. Use a structured approach to identify key variables that correlate with the issue.

Example Questions:

- Were there any app releases or updates around the time of the spike?
- Has driver availability changed? Are fewer drivers completing rides?
- Are cancellation rates different across geographies?
- Are cancellation rates higher for certain ride types (e.g., shared vs. private)?

At this stage, assume your interviewer confirms:

- No major app releases coincided with the spike.
- The number of active drivers has decreased.

Step 4: Refine Hypotheses

Now, use the gathered insights to refine or discard initial hypotheses.

Updated Hypothesis:

- Since there is no evidence of a technical issue or product change, the issue is likely operational or external.
- A drop in active drivers could mean that longer wait times are causing riders to cancel rides.

Check with the interviewer to validate your refined hypothesis before proceeding.

Step 5: Identify the Root Cause

Once sufficient evidence is gathered, summarize your findings concisely.

Example Conclusion:

"Based on the data, the root cause appears to be a decrease in the number of active drivers. This likely led to increased wait times, causing riders to cancel more frequently. The next step would be to investigate what led to this drop in driver participation—such as changes in incentives, external events, or operational issues."

Step 6: Evaluate and Recommend Solutions

Conclude by discussing the impact of the issue and proposing solutions.

Potential Solutions:

- Adjust driver incentives to improve participation.
- Provide ride discounts during peak times to reduce cancellations.
- Improve communication in the app regarding estimated wait times.
- Launch a driver outreach program to understand their concerns.

Common Pitfalls to Avoid

RCA questions test your ability to think critically, ask the right questions, and work with incomplete data. By following this structured approach, you can effectively diagnose problems and propose actionable solutions, positioning yourself as a strong candidate in PM interviews.

- **Jumping to Conclusions:** Avoid jumping to conclusions about the root cause too quickly. Focusing too much on proving or disproving a single hypothesis can slow down the process, making it take longer to find the right answer. This might lead to the interviewer stepping in to redirect the approach.
- **Not Explaining Thought Process:** Always articulate thoughts and explain the reasoning behind questions. Without this, it's easy to lose track of the approach, and the reasoning may seem unclear or disorganized to the interviewer. Clear communication keeps things structured and allows the interviewer to provide meaningful guidance.

Practise Question 1: SBM Churn Issue

Interviewer: Over the past three months, we have observed a 30% increase in customer churn, primarily among small to medium-sized business (SMB) clients. How would you approach solving this problem?

Interviewee: I had taken a structured approach to diagnosing and resolving this issue, following these six steps:

Step 1: Clarify and Gather Context

Interviewee: Before jumping to conclusions, I start by understanding the full scope of the problem. I have a few clarifying questions:

Understanding the Churn Trend:

- Has the churn rate increase been gradual or sudden over the past three months?
- Are SMB clients churning earlier in their lifecycle, or are long-time customers also leaving?

Segmentation of Affected Customers:

- Are there specific industries, geographies/business sizes within SMBs that are churning more?
- Are churning SMBs predominantly low-revenue or high-revenue clients?

Internal & External Factors:

- Have there been any internal changes, such as pricing updates, product modifications, or customer support shifts?
- Have competitors launched new features, price cuts, or marketing campaigns that might be influencing churn?

Churn Behaviour Analysis:

- Are customers switching to competitors/downgrading their plans/completely stopping usage?
- Have we collected exit survey data from recently churned customers?

Interviewer:

- Churn rate jumped suddenly in the last three months.
- It is highest among budget-conscious SMBs (annual revenue below ₹50 lakhs).
- We increased prices by 10% three months ago.
- A competitor launched a cheaper AI-powered product around the same time.
- Customer support waits times increased due to staff shortages.

Interviewee: Ok, now I will move to forming initial hypotheses.

Step 2: Form High-Level Hypotheses

Interviewee: I will formulate some high-level hypotheses about why SMB churn has increased.

- Pricing & Perceived Value – The 10% price increase may have made our product less attractive, especially for budget-conscious SMBs.
- Competitive Pressure – The new AI-powered competitor may be offering a better value proposition at a lower price.
- Customer Support Issues – Increased wait times may be frustrating SMBs, leading them to switch to competitors with better support.
- Product Shortcomings – Our feature set may no longer meet the evolving needs of SMBs.

- Macroeconomic Factors – SMBs might be cutting costs due to economic challenges, leading to lower discretionary spending on software.

Interviewer: Ok, how would you validate or eliminate these hypotheses?

Step 3: Gather Data

Interviewee: To test these hypotheses, I'd collect data from multiple sources:

- Customer Feedback & Exit Surveys – Reviewing recent churn survey responses to see if price, support, or competitors were common reasons for leaving.
- Support Ticket Analysis – Checking if there was a spike in support complaints about pricing, service delays, or missing features.
- Usage Data – Identifying if SMBs reduced usage before churning, possibly indicating lower perceived value.
- Competitive Analysis – Comparing our product's pricing and features with the new competitor's AI- powered solution.
- Churn Cohort Analysis – Breaking churned SMBs into groups based on industry, geography, and tenure to find common patterns.

Interviewer: Here is what we found:

- 70% of exit survey responses mentioned pricing as a key issue.
- 30% of churned SMBs specifically mentioned switching to AI-powered competitor.
- Support tickets increased by 25% in the past three months, with long wait times being a recurring complaint.

Interviewee: Ok, I will again refine my hypotheses.

Step 4: Refine Hypotheses and Repeat

Interviewee: Based on this data, I can eliminate the macroeconomic factors since customers are switching rather than cutting software spending entirely. I now believe the churn is primarily driven by three factors:

- Pricing Mismatch – SMBs don't see enough added value to justify the price hike.
- Competitive Threats – The cheaper AI-powered alternative is appealing to budget-conscious SMBs.
- Customer Support Delays – Poor customer support is frustrating SMBs, making them more likely to switch.

Interviewer: That aligns with our findings. What would you propose as a solution?

Step 5: Identify the Root Cause and Implement Solutions

Interviewee: To address each root cause, I'd propose a three-pronged strategy:

Issue	Solution	Implementation
Pricing Mismatch	Introduce tiered pricing to offer more affordable SMB plans.	Create a lower-cost SMB plan with limited features while maintaining profitability.
Competitive Pressure	Emphasize unique differentiators in marketing & launch a win-back campaign.	Offer one-time discounts or feature trials to regain lost customers.
Customer Support Delays	Improve response times with AI chatbots & additional agents.	Hire more customer support staff & implement self-service options for common issues.

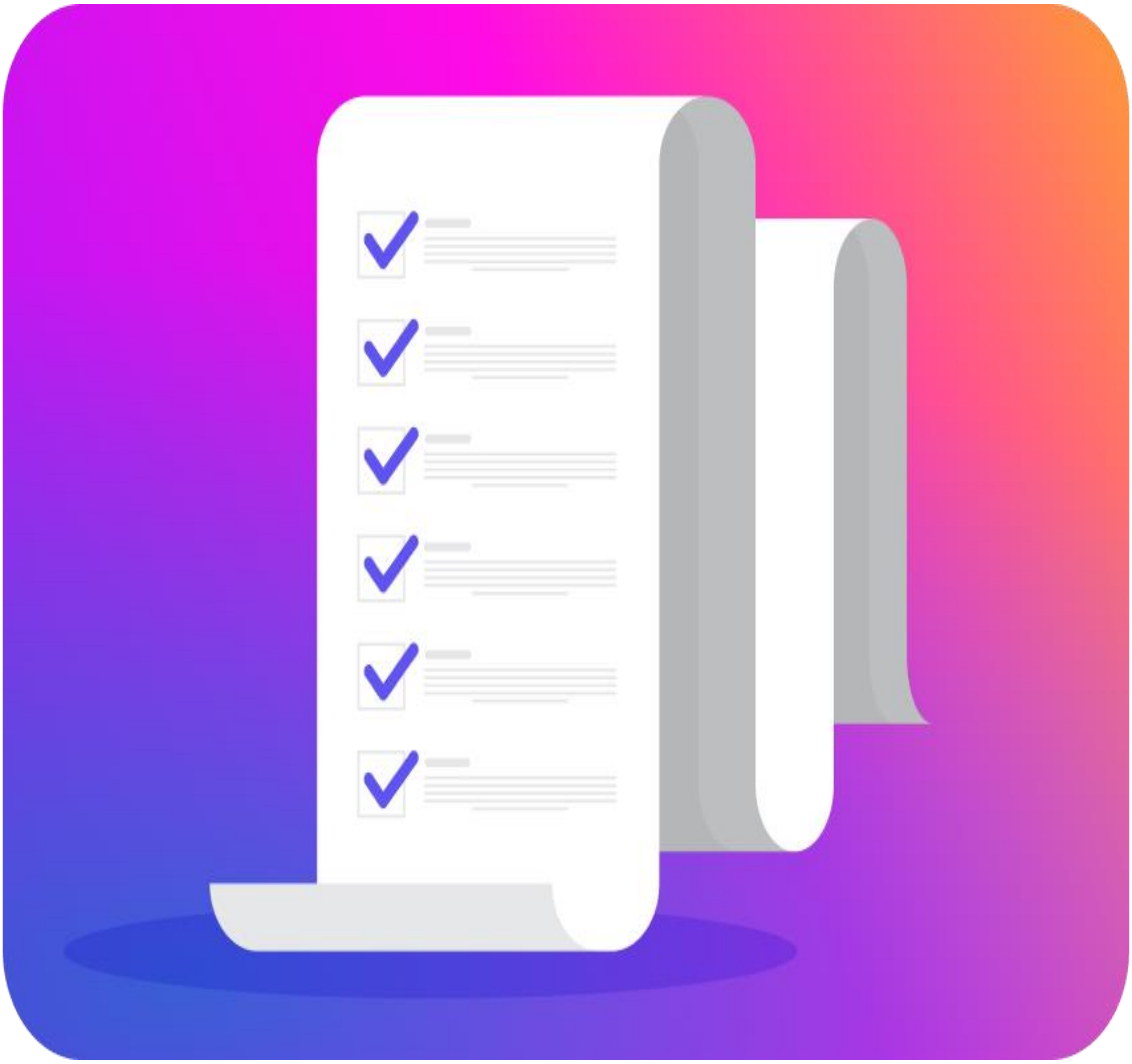
Interviewer: That is a solid approach. How would you track the effectiveness of these solutions?

Step 6: Measure Impact and Ensure Sustainability

Interviewee: To measure the success of these strategies, I'd track the following key metrics:

- Churn Rate Reduction – Monitor if the churn rate drops from 30% back to normal levels over the next 3-6 months.
- Customer Retention – Measure whether SMBs are staying longer after the pricing and support improvements.
- Customer Satisfaction (NPS & CSAT Scores) – Track if support ratings improve post-implementation.
- Win-Back Rate – Analyse how many previously churned SMBs return due to new pricing & offers.
- Revenue Impact – Ensure the new pricing structure does not significantly lower overall revenue
- while retaining customers.

Interviewer: Ok, Thank you.



Product Requirements Document

PRODUCT REQUIREMENTS DOCUMENT (PRD)

A Product Requirements Document (PRD) serves as a foundational document that outlines the problem being addressed, the market opportunity, the target users, the product that will solve the problem, key stakeholders, the sequence of tasks to be followed, and how the performance of the product will be measured. It acts as a bridge between different teams in an organization, ensuring alignment and collaboration.

As a product manager, my role is to take inputs from customers, engineering teams, design teams, and sales teams to ensure that everyone remains on the same page throughout the product development lifecycle.

Things to be included in PRD

PRD varies from project to project. However, a well-structured PRD should follow a cohesive story from beginning to end, with each section logically connected to the next. Below are the key components that should be covered in a PRD

Problem Definition: This section should provide a brief yet clear description of the problem that the product is trying to solve.

- Clearly define the pain points faced by users.
- Highlight any gaps in existing solutions.
- Address any assumptions or potential misconceptions upfront to prevent confusion later.

Market Research & Opportunity: This section should include:

- Research on existing competitors and alternative solutions.
- Market size and potential growth opportunities.
- Gaps in the market and how our product will address them.
- Customer demand trends and industry insights.

User Research: User research plays a crucial role in understanding the end users and their needs. This section should include:

- Primary and secondary research findings.
- User demographics, behaviour, and pain points.
- Data from surveys, focus groups, and in-depth interviews.
- Key insights derived from the research.

User Personas and Use Cases: Based on user research key user personas & their use cases are defined.

- A detailed description of the target users, including their roles, behaviours, and needs.
- Common scenarios and pain points faced by users.
- Use cases explaining how the product will address these pain points.

Prioritization of Use Cases: Once the use cases are identified, we prioritize them based on:

- Empathy maps to understand user emotions and frustrations.
- Data-backed reasoning for choosing specific use cases.
- Prioritization frameworks such as the RICE method (Reach, Impact, Confidence, and Effort) or the MoSCoW method (Must-have, Should-have, Could-have, Won't-have).

Proposed Solutions: This section details the features and functionalities of the product, including:

- Functional requirements - What the product will do.
- Non-functional requirements - Performance, security, scalability, and compliance factors.
- Clear explanations of how the product addresses the problem.
- Key differentiators from existing solutions.

Prioritization of Solutions: Not all solutions can be implemented at once, so prioritization is key:

- Impact vs. effort analysis to determine quick wins and long-term goals.
- Prioritization models such as the Kano Model (categorizing features into basic needs, performance needs, and delighters) or the MoSCoW method.
- A phased approach to implementation, ensuring maximum impact in early releases.

UX Design: To provide a visual representation of the product, this section includes:

- Wireframes created using tools like Figma or Balsamiq.
- User flows and navigation structures.
- Design principles that enhance user experience.

Roadmap: The roadmap provides a clear plan for product development, ensuring alignment across teams. The key steps include:

Step 1: Idea Sourcing

Step 2: Idea Screening

Step 3: Market Research & User Research

Step 4: Strategy Development.

Step 5: Product Creation

Step 6: Testing & Feedback Gathering

Step 7: Focus on Product Improvement

Product Design

PRODUCT DESIGN

Product design questions are used to test a candidate's creativity, problem-solving ability and understanding of user needs. These questions should be addressed in a highly structured way, even if they appear to be open ended. Prioritizing the understanding of user needs is crucial in this context. They primarily focus on either enhancing an existing product or conceptualizing a new one. These products can be physical or digital.

Examples of Product Design Questions:

You are a PM at Uber; how would you increase driver retention

Design a smart backpack for students.

Conceptualize an AI-powered kitchen appliance that simplifies cooking for beginners.

Design a self-cleaning water bottle

How would you improve the user experience of Google Maps?

Design a next-generation smartwatch for fitness enthusiasts.

Conceptualize an AI-powered virtual study companion for students.

Design a self-adjusting ergonomic office chair for long working hours.

How would you enhance the user experience of a mobile banking app?

Practise Question 1: Product Development

Interviewer: Design a smart backpack for students.

Candidate: Before diving into the design, I'd like to consider the different types of students, their unique needs, and how we can create a differentiated product. Broadly, students can be grouped into the following categories:

1. School Students (K-12) – Need a lightweight, organized, and safe backpack.
2. College & University Students – Require tech-friendly features like charging, security, and efficient storage.
3. Commuting Students – Need anti-theft features, weatherproofing, and ergonomic support for long travel.

When thinking about a smart backpack, some core functionalities could include:

- Organized storage and accessibility.
- Built-in charging for electronic devices.

- Anti-theft and tracking features.
- Weight distribution for better comfort.

Now, I'd like to narrow down the best fit between the type of student and the most impactful smart features.

School students typically don't need extensive tech integration, and commuter students primarily value durability and security. However, college and university students rely heavily on technology, carry multiple devices and spend long hours on campus, making them the ideal target for a smart backpack.

Product Mission: To create a smart backpack for college students that enhances organization & security.

We can segment students further based on their daily routines and requirements:

1. Tech-Savvy Students – Carry multiple gadgets, need power on the go.
2. Library & Study-Oriented Students – Require easy organization and accessibility.
3. Commuters – Need theft protection and comfort for long-distance travel.
4. Athletes & Active Students – Need lightweight durability for movement.

Since our mission focuses on connectivity and organization, I will prioritize tech-savvy students, as they would benefit the most from a feature-rich smart backpack.

Can I proceed with this user segment?

Interviewer: Go ahead!

Candidate: User Journey & Key Needs:

1. Morning Preparation – Efficient packing, ensuring nothing is forgotten.
2. Commute to Campus – Charging devices, quick access to essentials.
3. Attending Classes – Keeping everything organized and easily accessible.
4. Study Sessions & Breaks – Secure storage for when the backpack is left unattended.
5. Heading Back Home – Comfort and weight distribution after a long day.

Potential Smart Features to Address These Needs:

1. Integrated Charging System – Built-in power bank with wireless and USB-C charging.
2. Smart Organization System – RFID-tagged compartments for tracking lost or forgotten items.
3. Anti-Theft & Security Features – GPS tracking, fingerprint lock, and motion detection.
4. Ergonomic Comfort Enhancements – Weight distribution sensors and memory foam straps.

To ensure effective prioritization, I will analyse features based on impact on user experience versus effort required for development.

Feature	Impact	Complexity
Smart charging system	High	Medium
RFID-powered compartment tracking	High	High
Anti-theft security (GPS & lock)	High	High
Ergonomic weight distribution	Medium	Low

Since power availability and security are the top pain points for tech-savvy students, I will focus on "Integrated Charging System & Anti-Theft Features" as our key differentiators.

Solution Details:

1. Built-in Smart Charging – Power bank with multiple ports, wireless charging.
2. Item Tracking & Organization – RFID integration to alert users if an important item (like a laptop) is missing.
3. Security Features – Bluetooth-enabled tracking, app alerts for unauthorized movement, and a fingerprint lock for valuables.
4. Companion Mobile App – Allows students to track items, check battery levels, and receive theft alerts.

Interviewer: Very well structured. Thank you

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Solved Example 2: Product Improvement

Interviewer: As a Product Manager, how would you reduce subscription cancellations for Netflix?

Candidate: I'd like to break down churn by identifying the key reasons why users leave and the different types of subscribers we need to consider. Types of Subscription Cancellations:

1. Voluntary Churn – Users actively cancel due to dissatisfaction, pricing concerns, or content fatigue.
2. Involuntary Churn – Cancellations caused by payment failures or forgotten renewals.

Potential Reasons for Churn:

- Content-related issues (Lack of engaging content, not enough variety, missing regional preferences).
- Pricing concerns (Users finding it expensive compared to competitors).
- User experience problems (Difficulty in discovering relevant content).

- Temporary users (Subscribers who only join for a specific show and cancel afterward).
- Payment failures (Credit card expiration, insufficient funds, or technical issues).

To narrow my focus, I'd prioritize tackling voluntary churn, as it represents users who actively choose to leave and can be retained with the right strategies.

Product Mission: *To enhance Netflix's user experience and engagement, ensuring long-term retention by addressing content discoverability, pricing perception and user satisfaction.*

***I*nterviewer:** Who are the key users and how do their needs impact cancellations?

Candidate: Based on different user behaviours, we can segment them into:

1. Casual Viewers – Watch infrequently, may not find enough value to continue.
2. Binge Watchers – Subscribe for a specific show, then cancel after finishing it.
3. Budget-Conscious Users – Sensitive to price changes and compare with competitors.
4. Family & Shared Account Users – Look for multi-user benefits and parental controls.
5. Loyal Long-Term Subscribers – Less likely to cancel but may disengage over time.

Since our goal is to reduce churn, I will focus on Casual Viewers & Binge Watchers, as they have the highest risk of cancellation due to content fatigue or lack of engagement.

Can I proceed with this user segment?

Interviewer: Go ahead!

Candidate: User Journey & Pain Points Leading to Cancellation:

1. On boarding & First Experience – Struggle to find engaging content, leading to low initial attachment.
2. Regular Usage – Content fatigue, lack of personalized recommendations, or dissatisfaction with library updates.
3. Subscription Renewal Decision – Perceived lack of value compared to competitors.
4. Cancellation Process – No effective retention measures before final cancellation.

Potential Solutions to Address These Pain Points:

1. Personalized Content Engagement – AI-driven recommendations based on unfinished shows, past watch history, and social trends.
2. Loyalty & Reward System – Discounts or benefits for long-term subscribers, gamifying engagement.
3. Better Pricing Options – Flexible plans (e.g., per-show rentals, student discounts).
4. Reactivation Strategies – Smart win-back campaigns offering discounts or exclusive content.

To ensure effective prioritization, I will analyse features based on impact on retention vs. development complexity.

Feature	Impact	Complexity
AI-driven Engagement Features	High	High
Loyalty & Rewards Program	High	Medium
Flexible Pricing Options	Medium	Medium
Win-back Campaigns	Medium	Low

Since personalized content engagement & loyalty rewards provide the highest impact on retention, I will focus on these solutions.

Solution Details:

1. Enhanced Personalized Recommendations – AI-driven content curation that adjusts in real time based on watching patterns, unfinished series, and trending genres.
2. "Stay Longer, Pay Less" Model – Introduce loyalty-based discounts for long-term subscribers, reducing churn caused by pricing concerns.
3. Engagement Triggers – Timely reminders to users who haven't watched anything recently, nudging them back with curated suggestions.
4. Smart Cancellation Flow – If a user attempts to cancel, offer a "pause subscription" option instead of cancelling outright, allowing them to return seamlessly later.

Interviewer: Very well structured. That will be all.

Common Pitfalls in Product Design Questions

Jumping straight to features

Diving into feature ideas without first understanding user needs or defining the problem leads to an incomplete answer.

Disorganized responses

A scattered thought process results in an unstructured response, making it harder for the interviewer to follow.

Lack of clarification

If the question seems ambiguous, failing to ask clarifying questions can lead to misinterpretation and irrelevant solutions.

Behavioural Questions

BEHAVIOURAL QUESTIONS

People tend to skip this section a lot during their preparation phase but it quite important as one question usually comes from here during the interviews. So, it is better to be prepared in advance with the answers. Below are certain guidelines that will give your answer a proper structure and direction.

STAR Framework

Source: Springboard Success (LinkedIn)

This is a framework often followed by interviewees to answer the behavioural questions in a structured manner. The main ingredient is to utilize real life examples in the framework

Example:

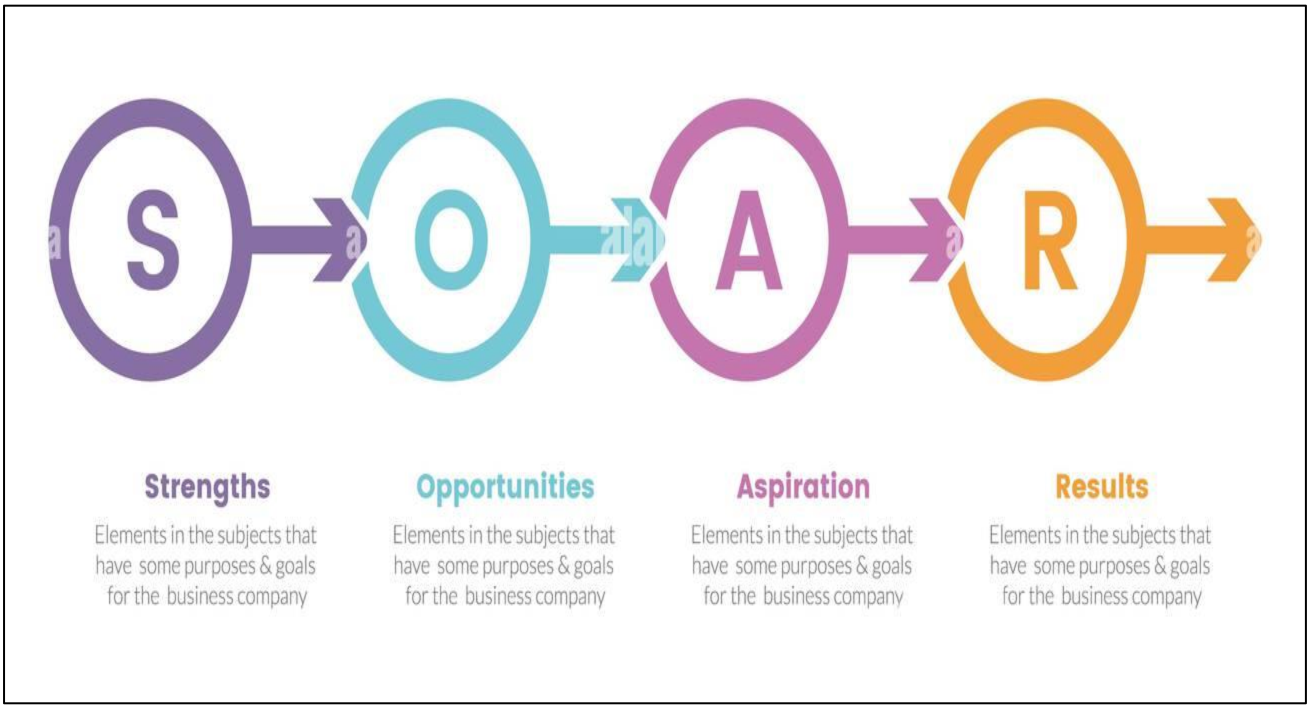
"Can you share an experience where you had to persuade a team or stakeholder to adopt a new idea or process?"

Response:

- **Situation** – Set the context. What was the situation? What type of idea was it? Who were the key stakeholders? Were there any initial concerns? Providing a such clear picture helps the interviewer understand the challenge.

- **Task** – Now that you had that situation what were you expected to do? Was there challenges you faced? Any plan you had to solve them? Tell what you needed to achieve to complete the task like getting buy-in, securing approval, or ensuring smooth implementation.
- **Action** – Once you have planned all the tasks, how did you execute it? Any hurdles while executing the plan? Did something go wrong during the process? Describe any tools you used during this step as well
- **Result** – This should be the easiest part as you should choose a situation with a favorable result for you. Just tell how it helped all your stakeholders and organization. And try to quantify it as well if possible.

SOAR Framework



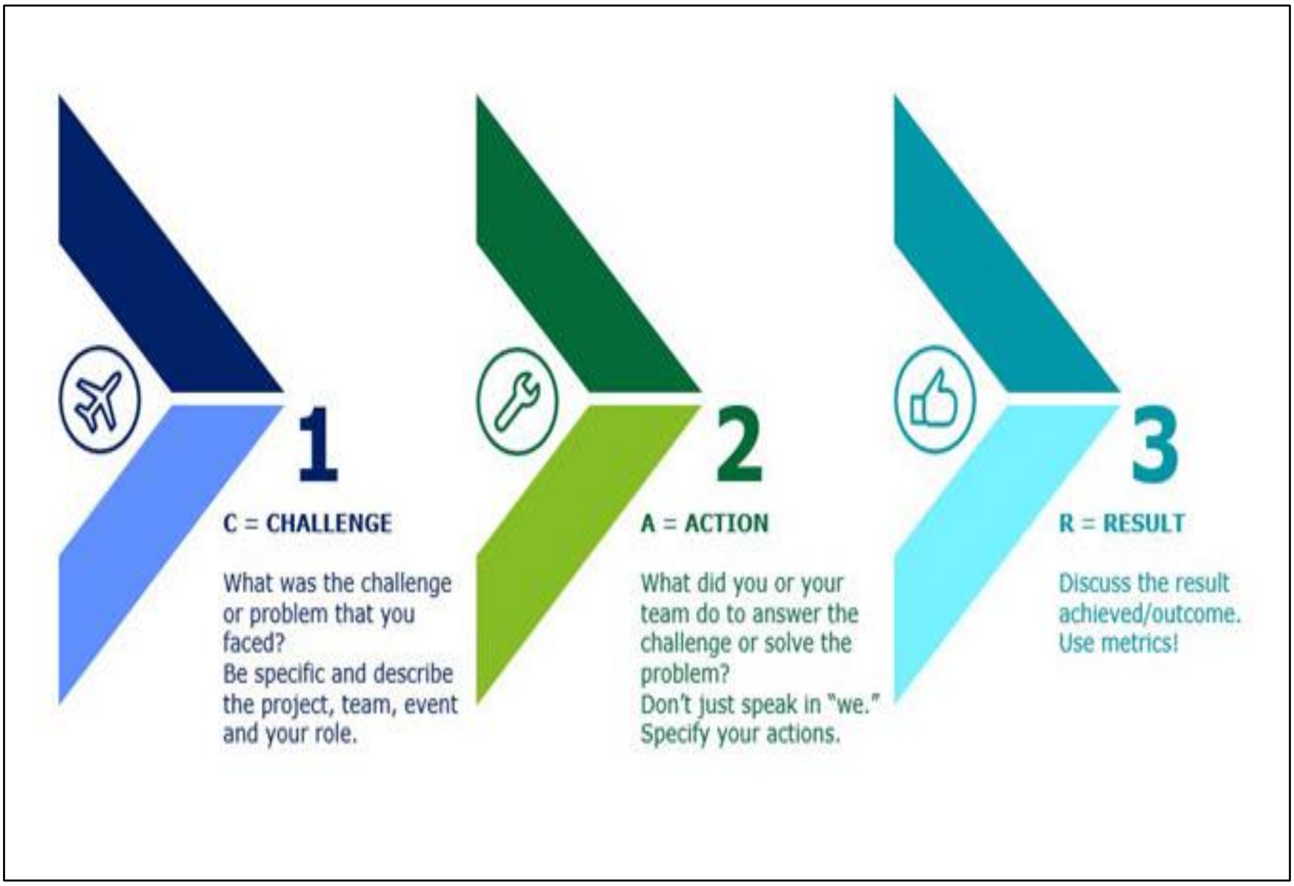
Source: Alamy

Exactly same as STAR framework but just instead of task we use "Obstacles", where we talk about the different challenges and hurdles, we faced during our approach and how we were able to solve them.

Why Use SOAR?

- Demonstrates resilience – Shows how you navigate setbacks and problem-solve effectively.
- Makes answers more compelling – Interviewers get a clear picture of how you handle real-world challenges.
- Highlights adaptability – Showcases your ability to think on your feet and adjust to changing circumstances.

CAR Framework



Source: Deloitte

- **The Challenge** – Think of a time when you faced a tough situation or had an important task with obstacles in the way. What was the problem? Why was it difficult? Give enough details so the interviewer understands what you were dealing with.
- **What You Did** – Focus on your role in solving the problem. Even if you worked with a team, explain what you did to make a difference. What did you come up with? What didn't you do?
- **The Outcome** – How did things turn out? Did you solve the problem, improve the situation, or learn something valuable? Even if things didn't go perfectly, what was the result, and how did it help you or others?

QUESTIONS TO PREPARE FOR INTERVIEW

Why do you want to be a Product Manager?

Different companies expect different things from a PM, so first, learn what this company looks for. Think about why you like this role—do you enjoy solving problems, working with different teams, or making things better for users? Connect it to your own skills and experiences. If you say you are a big-picture thinker, give a real-life example where your ideas made a difference.

What if you don't get a PM job?

Don't say "I don't know" or "I never thought about it." PMs need to plan ahead, so show them you have a backup plan. Maybe you'll gain more experience and try again later, or maybe you're also interested in roles like marketing or business strategy. Show that you're open to learning and growing.

You do not have IT experience—why do you think you can be a PM?

A lot of freshers worry about this. If you haven't worked in tech, talk about what you have done. Have you joined case competitions, built a project, or taken online courses? Show that you are eager to learn. Also, highlight skills like problem-solving, teamwork, and decision-making — these are just as important for a PM.

Tell me about a time you had a problem in a team. How did you fix it?

Think of a time you worked in a group—maybe for college, an internship, or an event. What was the issue? Did people disagree? Was there a missed deadline? A lack of coordination? Explain how you helped solve the problem. Also, share what you learned and how you used that lesson later.

What's more important — speed or quality?

There is no one right answer. Sometimes, getting a product out fast is key (like launching a test version). Other times, quality matters more (like in healthcare or banking). Don't just pick one—explain when each one is the better choice. If you can, ask the interviewer for a situation before answering.

Tell me about a time you used data to convince someone.

PMs use data to make decisions. Have you ever used numbers to prove a point? Maybe you showed survey results to improve a project or used feedback to suggest a change. If you know tools like Excel, Tableau, or Power BI, mention that too!

Why do you want to work for this company?

Don't give a generic answer. Do your homework—learn about their vision, products, and culture. What excites you about them? Why do you think you fit in? Be specific and show that you really care.

How do you decide what tasks are most important?

PMs must choose what to work on first. How do you decide? Do you look at deadlines, business needs, or customer impact? Explain your method in simple steps (like making a list, ranking by urgency, or using a scoring system).

What is one thing you would improve in our product?

This is where your research helps. Think like a user but also consider what is possible for the business and tech teams. Don't just point out a problem, explain how your idea would help and why it makes sense.

Note: These are not the only questions you might get. The interviewer can ask more based on what you say. So, always be ready to explain your answers and anything on your resume.

You might also get some general questions that are not about Product Management, like:

- Tell me about yourself.
- What are your strengths and weaknesses?
- How do you handle mistakes?
- Do you think you will fit into our company culture?
- What is something unique about you?
- What are your hobbies and interests?

Thank You for Reading, Hope it was Helpful :)